

Physics of Complex Systems

Stochastic processes and nonequilibrium statistical physics — Lecture Notes

Alvaro Mendez

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About these notes

Scope

These are lecture notes for *Physics of Complex Systems* (third-semester course, Master in Physics of Data, University of Padova). A **complex system** is one whose large-scale behaviour cannot be read off the equations of its individual constituents: it *emerges* from the collective dynamics of very many degrees of freedom, most of which we can neither follow nor control. The natural language for such systems is that of **stochastic processes** and **nonequilibrium statistical physics** — we give up on individual trajectories and describe instead how *probabilities* evolve in time.

The course moves through the following topics:

1. Brownian motion, stochastic differential equations and stochastic processes (Langevin and Fokker–Planck descriptions).
2. Markov chains and the detailed-balance condition; equilibrium and nonequilibrium steady states (NESS).
3. Linear response theory and transport; dynamic susceptibility, the fluctuation–dissipation theorem and the Kramers–Kronig relations.
4. Onsager theory of coupled transport (reciprocity, Seebeck and Peltier effects); fluctuation theorems, work identities, generalised detailed balance and entropy production.
5. Nonequilibrium phase transitions: absorbing phases and directed percolation; driven systems and the asymmetric simple exclusion process.
6. Information theory and the data-transmission problem: Shannon’s coding theorem, the random energy model and random code ensembles.

Reading the notes

The notes are written lecture by lecture and grow with the course.

- **Part I (Chapters 1–2):** from the microscopic picture of a particle kicked by molecules to the probabilistic description of diffusion. Chapter 1 builds the random walk, Einstein’s theory of Brownian motion and the Langevin equation, and sets up the master equation. Chapter 2 derives the Fokker–Planck equation, reads it as a continuity equation for probability, distinguishes stationarity from equilibrium, and closes with the Chapman–Kolmogorov equation and its forward/backward Kolmogorov forms.

Every chapter uses the styled callout environments to separate definitions, results, worked examples and remarks from the surrounding prose, and ends with exercises (solutions collapsible in the HTML version) and a *Sources* section. Diagrams are used wherever they make the physics visible: separation of time scales, balance of currents, spreading of probability, gain and loss of probability in the master equation.

Notation and conventions

- **Temperature in energy units:** $k_B = 1$ throughout, so T is measured in joules and $k_B T \rightarrow T$.

- $\langle \cdot \rangle$: average over realisations of the noise (equivalently, over an ensemble of replicas).
- $\tilde{\gamma}$: friction coefficient (force per unit velocity), $\tilde{\gamma} = 6\pi\eta R$ for a sphere in the Stokes regime; $\gamma = \tilde{\gamma}/m = 1/t_d$ the corresponding relaxation rate.
- $\tilde{\eta}(t)$: stochastic force; $\eta(t) = \tilde{\eta}(t)/m$ the stochastic force per unit mass; noise strengths $\tilde{\Gamma}$ and $\Gamma = \tilde{\Gamma}/m^2$.
- D : diffusion coefficient; d : spatial dimension.
- $p(\mathbf{x}, t)$: probability density of the state $\mathbf{x}$ at time t .
- $R(\mathbf{x}, \mathbf{x}')$: **transition rate from $\mathbf{x}'$ to $\mathbf{x}$** (probability per unit time and per unit volume). *Warning:* Livi–Politi write $W(\mathbf{x}, \mathbf{x}')$ for the same object in §1.4.2; we use R from the start, consistently with the continuous-time master equation of §1.6, where R_{ik} is the rate from state k to state i .
- $W(x_0, t_0 | x, t)$: transition probability density to be at x at time t given x_0 at time t_0 (time flows from left to right).
- $\mathbf{J}(\mathbf{x}, t)$: probability current.

Sources

The course follows

- Roberto Livi and Paolo Politi, *Nonequilibrium Statistical Physics: A Modern Perspective*, Cambridge University Press, 2017 (Chapters 1 and following);

and, for the information-theory part,

- Marc Mézard and Andrea Montanari, *Information, Physics, and Computation*, Oxford University Press, 2009.

Every chapter closes with its own *Sources* section naming the specific lectures and book sections it draws on.

Part I.

Part I: Brownian motion and stochastic
processes

1. Random walks, Brownian motion and the Langevin equation

A grain of pollen in water jiggles forever, along a path so irregular that no two observations ever repeat. Behind that jiggling lies one of the founding ideas of nonequilibrium statistical physics: the *macroscopic* laws of transport (diffusion, friction, viscosity) emerge from an enormous number of *microscopic* collisions that we can neither follow nor predict, and the only sensible description is a statistical one. This chapter builds that description step by step. We start from the kinetic theory of a dilute gas (how often do molecules collide, and how far do they fly?), turn the sequence of collisions into a **random walk** and find the diffusive law $\langle X^2 \rangle \propto t$. We then follow Einstein's argument for **Brownian motion**, whose key ingredient is a *separation of time scales*, and Langevin's reformulation as a Newton equation with a random force, which yields the first example of a **fluctuation–dissipation relation**. We close by changing perspective — from trajectories to probabilities — and writing the **master equation** that Chapter 2 will turn into the Fokker–Planck equation.

1.1. Complexity, emergence and the need for statistics

A mole of gas contains $N_A \simeq 6.022 \times 10^{23}$ molecules. Even if Newton's laws for each of them were known exactly, the initial conditions are not, and tiny uncertainties are amplified by every collision. Yet the *collective* behaviour is extremely regular: pressure, temperature, viscosity and diffusion constants are reproducible numbers. This is the archetype of an **emergent property**: a law that holds at the macroscopic scale, is insensitive to most microscopic details, and is invisible in the equations of a single constituent.

Two ideas recur throughout the course.

- **Separation of scales.** Microscopic events (collisions) happen on time and length scales much shorter than the ones on which we observe the system. Averaging over the fast scales leaves an effective, often *stochastic*, dynamics for the slow ones.
- **From trajectories to distributions.** Following Boltzmann, instead of individual trajectories we describe the evolution of a **probability distribution**. Randomness is not a statement about the underlying mechanics, but an effective (and remarkably successful) way of encoding everything we choose not to resolve.

Historically, these ideas met fierce resistance: they presupposed the atomic hypothesis when there was no direct evidence for atoms, and they seemed to clash with the time-reversibility of mechanics. Brownian motion — and Perrin's measurements built on Einstein's theory — is precisely what settled the question of the existence of atoms.

1.2. Kinetic theory of the ideal gas

The simplest mechanical model of a thermodynamic system is Boltzmann's **ideal gas**: N identical hard spheres of mass m and radius r in a volume V , with number density $n = N/V$. The gas is **dilute**: the

1. Random walks, Brownian motion and the Langevin equation

typical inter-particle distance is much larger than the particle size,

$$\delta = n^{-1/3} \gg r.$$

For hydrogen at room temperature and atmospheric pressure, $\delta \sim 10^{-8}$ m while $r \sim 10^{-10}$ m. Between collisions particles move on straight lines at constant velocity; collisions are binary (triple collisions are negligibly rare) and elastic (momentum and kinetic energy are conserved).

1.2.1. Collision rate and mean free path

How often does a given molecule collide? A sphere of radius r moving at velocity $\mathbf{v}$ hits every other sphere whose centre comes within a distance $2r$ of its own centre. In a time t it therefore sweeps a cylinder of base

$$\sigma = \pi(2r)^2 = 4\pi r^2 \quad (\text{collision cross section})$$

and height $|\mathbf{v}|t$. Treating the targets as momentarily at rest, the number of collisions is the number of centres in that cylinder:

$$N_{\text{coll}} = n \sigma |\mathbf{v}| t.$$

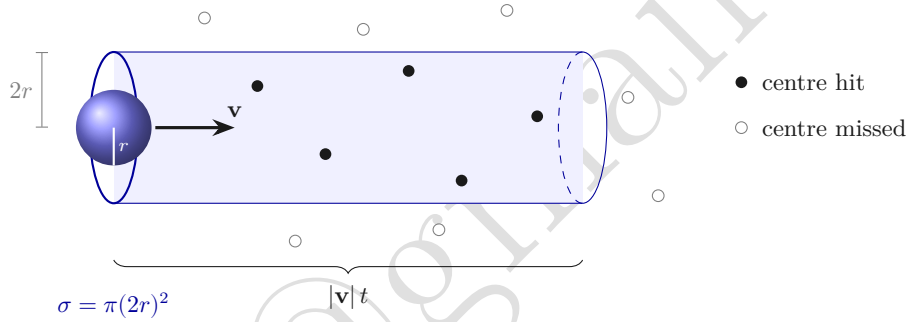


Figure: the collision cylinder. A molecule hits every other molecule whose centre lies within $2r$ of its path, i.e. inside the cylinder of cross section $\sigma = 4\pi r^2$ swept in time t . The expected number of hits is $n\sigma|\mathbf{v}|t$.

The collision rate is $N_{\text{coll}}/t = n\sigma|\mathbf{v}|$, and the mean time between collisions is its inverse. To get a number, replace $|\mathbf{v}|$ with its thermal average $\langle v \rangle$ computed with the **Maxwell distribution** of speeds,

$$P(v) = \frac{4}{\sqrt{\pi}} \left(\frac{m}{2T} \right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2T}\right),$$

whose three characteristic speeds are

$$v_{\text{max}} = \sqrt{\frac{2T}{m}}, \quad \langle v \rangle = \sqrt{\frac{8T}{\pi m}} = \frac{2}{\sqrt{\pi}} v_{\text{max}}, \quad \langle v^2 \rangle^{1/2} = \sqrt{\frac{3T}{m}} = \sqrt{\frac{3}{2}} v_{\text{max}}.$$

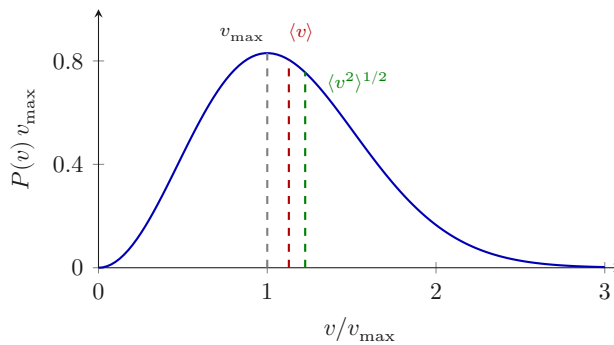


Figure: the Maxwell distribution of molecular speeds, with $v_{\max} < \langle v \rangle < \langle v^2 \rangle^{1/2}$. The v^2 factor is the phase-space volume of a spherical shell; the Gaussian factor is the Boltzmann weight of the kinetic energy.

Definition 1.1 (Collision time and mean free path). The **mean collision time** τ and the **mean free path** λ (average distance flown between two successive collisions) are, in the approximation of immobile targets,

$$\tau = \frac{1}{n\sigma\langle v \rangle}, \quad \lambda = \langle v \rangle \tau = \frac{1}{n\sigma}.$$

The targets are not at rest, however. What matters is the **relative velocity** $\mathbf{v}_r = \mathbf{v}_1 - \mathbf{v}_2$ of the two colliding molecules. Each Cartesian component of $\mathbf{v}_{1,2}$ is Gaussian with variance T/m , and the difference of two independent Gaussians is Gaussian with the *sum* of the variances, $2T/m$. Hence $\mathbf{v}_r$ is Maxwell-distributed with $m \rightarrow m/2$,

$$P_r(v_r) = \frac{4}{\sqrt{\pi}} \left(\frac{m}{4T} \right)^{3/2} v_r^2 \exp\left(-\frac{mv_r^2}{4T}\right), \quad \langle v_r \rangle = \sqrt{\frac{16T}{\pi m}} = \sqrt{2} \langle v \rangle.$$

Replacing $\langle v \rangle$ by $\langle v_r \rangle$ in the collision rate gives the improved estimates

$$\tau = \frac{1}{\sqrt{2}n\sigma\langle v \rangle}, \quad \lambda = \langle v \rangle \tau = \frac{1}{\sqrt{2}n\sigma}.$$

Note that $\lambda/\tau = \langle v \rangle$, not $\langle v_r \rangle$: the distance travelled *by one particle* in time τ is set by its own speed.

Remark (Orders of magnitude). For a gas at room temperature and pressure, $\lambda = \mathcal{O}(10^{-7} \text{ m})$, three orders of magnitude larger than the molecular size $r = \mathcal{O}(10^{-10} \text{ m})$, and $\tau = \mathcal{O}(10^{-10} - 10^{-11} \text{ s})$. The hierarchy $r \ll \delta \ll \lambda$ is what makes the gas “ideal”: molecules are point-like on the scale of their separation, which in turn is small compared with the distance they fly freely.

1.3. The random walk: a basic model of diffusion

Fix your attention on one molecule of a gas in equilibrium at temperature T . Each collision deflects it, and its trajectory becomes a zig-zag of straight segments. Predicting that zig-zag with Newton’s laws would require the velocities of all the molecules it will ever meet — hopeless. Instead we **model** the collisions as *random events*.

Definition 1.2 (Random walk model of a gas molecule). A particle starts at the origin at $t = 0$. Between collisions it moves with constant speed v along straight segments $\mathbf{x}_i$ of fixed length $|\mathbf{x}_i| = \ell$. After each collision the new direction is **uniformly distributed** over the solid angle and **uncorrelated** with all previous ones. After N collisions the position is

$$\mathbf{X} = \sum_{i=1}^N \mathbf{x}_i.$$

The assumption of uncorrelated directions is where probability enters a purely mechanical problem. It cannot be derived from Newton’s laws; its justification is its success (and, physically, the sensitivity of collisions to the finite size and internal degrees of freedom of real molecules).

Mean displacement. By symmetry each segment averages to zero, $\langle \mathbf{x}_i \rangle = 0$, hence $\langle \mathbf{X} \rangle = 0$: there is no preferred direction.

1. Random walks, Brownian motion and the Langevin equation

Mean square displacement. Expanding the square,

$$\mathbf{X}^2 = \sum_{i=1}^N \sum_{j=1}^N \mathbf{x}_i \cdot \mathbf{x}_j = \ell^2 \sum_{i,j} \cos \theta_{ij} = \ell^2 \left(N + \sum_i \sum_{j \neq i} \cos \theta_{ij} \right),$$

where θ_{ij} is the angle between segments i and j and we separated the N diagonal terms ($\theta_{ii} = 0$). For $i \neq j$ the directions are independent and $\cos \theta_{ij}$ is a random number in $[-1, 1]$ with zero mean. Averaging over many **replicas** of the walk (independent realisations of the same process), the cross terms vanish:

$$\langle \mathbf{X}^2 \rangle = \ell^2 N.$$

The typical distance from the origin grows as $\sqrt{N}$, much more slowly than the total path length $N\ell$. The square root comes entirely from the *diagonal* terms: the walker forgets its direction at every step, so only the “self-correlation” of each step survives.

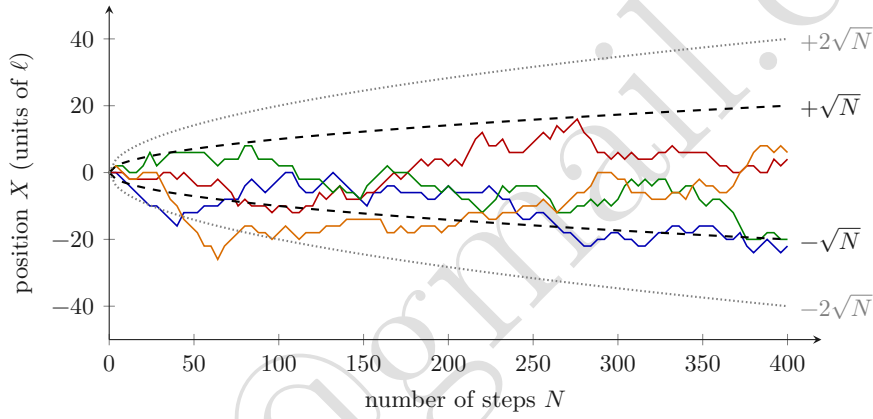


Figure: four independent replicas of a one-dimensional random walk with $\ell = 1$. Each trajectory is erratic and different, but the ensemble is confined, with high probability, within an envelope $\pm\sqrt{N}$ (one standard deviation, dashed) or $\pm 2\sqrt{N}$ (dotted). Statistical regularity emerges only at the level of the ensemble.

Example 1.1 (One-dimensional lattice walk and the Gaussian limit)

In $d = 1$ with steps $\pm\ell$ of equal probability, the position after N steps is $X = (2k - N)\ell$, where k is the number of steps to the right, a binomial variable:

$$P(X = (2k - N)\ell) = \binom{N}{k} 2^{-N}.$$

Its mean is 0 and its variance $N\ell^2$, in agreement with the general formula. For large N , by the central limit theorem (or Stirling’s formula) the distribution becomes Gaussian,

$$P(X) \simeq \frac{1}{\sqrt{2\pi N\ell^2}} \exp\left(-\frac{X^2}{2N\ell^2}\right).$$

With $N = t/\tau$ steps this is $\exp[-X^2/(4Dt)]/\sqrt{4\pi Dt}$ with $D = \ell^2/(2\tau)$ — exactly the solution of the diffusion equation that we shall derive in Chapter 2.

1.3.1. Random run lengths

Real flights are not all of the same length. Let $\mathbf{x}_i = \ell_i \hat{\mathbf{x}}_i$ with the lengths ℓ_i drawn independently from a normalised distribution $g(\ell)$ and the directions $\hat{\mathbf{x}}_i$ independent of everything else. Then

$$\langle \mathbf{x}_i \cdot \mathbf{x}_j \rangle = \langle \ell_i \ell_j \rangle \langle \hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j \rangle = \langle \ell^2 \rangle \delta_{ij} \implies \langle \mathbf{X}^2 \rangle = \langle \ell^2 \rangle N.$$

If collisions are independent events in time — a Poisson process — the free path is **exponentially distributed** with mean λ ,

$$g(\ell) = \frac{1}{\lambda} e^{-\ell/\lambda}, \quad \langle \ell \rangle = \lambda, \quad \langle \ell^2 \rangle = 2\lambda^2,$$

so that $\langle \mathbf{X}^2 \rangle = 2\lambda^2 N$. After a time t the particle has flown a total length $L = N\lambda = \langle v \rangle t$, hence

$$\langle \mathbf{X}^2(t) \rangle = 2\lambda \langle v \rangle t \equiv 2d D t, \quad D = \frac{\lambda \langle v \rangle}{d}.$$

Definition 1.3 (Diffusion coefficient). A process whose mean square displacement grows linearly in time is **diffusive**. In d spatial dimensions the **diffusion coefficient** D is defined by

$$\langle (\mathbf{X}(t) - \mathbf{X}(0))^2 \rangle \simeq 2d D t \quad (t \rightarrow \infty),$$

so that each Cartesian component contributes $2Dt$. For the random walk of a gas molecule, $D = \lambda \langle v \rangle / d$.

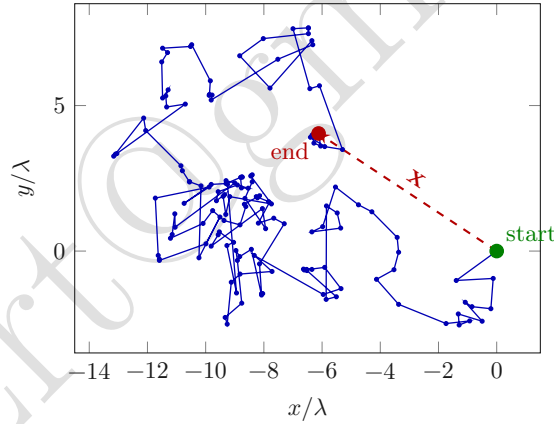


Figure: a two-dimensional random walk with exponentially distributed run lengths (mean λ) and uniformly random directions, 150 collisions. It looks like Perrin's microscope tracings of colloidal particles. The dashed arrow is the end-to-end displacement $\mathbf{X}$; its length fluctuates from walk to walk around the typical value $\sqrt{\langle X^2 \rangle} = \sqrt{2N} \lambda \simeq 17\lambda$ — only the ensemble average is fixed.

Remark (Diffusion is slow). For air at 20°C, $\langle v \rangle \sim 450$ m/s and $\lambda \sim 0.06 \mu\text{m}$, so $2\lambda \langle v \rangle \simeq 5 \times 10^{-5} \text{ m}^2/\text{s}$. Reaching $\sqrt{\langle X^2 \rangle} = 1$ m takes $t \simeq 2 \times 10^4$ s ≈ 5 hours, and 10 m takes about 20 days. Molecules are fast, but they waste almost all their motion going back and forth. In everyday life smells spread by convection and turbulence, not by diffusion.

The same kinetic argument (transport of a quantity over a distance λ by particles crossing a surface) gives Fick's law $J = -D \partial_z n$ with $D = \frac{1}{3} \lambda \langle v \rangle$ in $d = 3$, and analogous expressions for the viscosity $\eta = \frac{1}{3} n m \langle v \rangle \lambda$ and the thermal conductivity $\kappa = \frac{1}{3} \rho C_V \langle v \rangle \lambda$. All transport coefficients are fixed by a handful of microscopic quantities (m , $\langle v \rangle$, λ), and the random-walk result matches the transport one.

1.4. Brownian motion

In 1827 the botanist Robert Brown observed that pollen grains (size $\sim 10^{-3}$ cm) suspended in water move erratically and incessantly, more vigorously the smaller the grain. For almost a century the phenomenon was a puzzle. A naive kinetic argument even seemed to *rule out* a molecular explanation:

- a pollen grain is enormously heavier than a water molecule, so a single collision changes its velocity by a negligible amount ($\sim 10^{-6}$ m/s);
- collisions are extremely frequent and, by isotropy, equally likely from every direction, so on observable time scales their effects should *average out* to zero.

The flaw in this argument is that the *velocity* kicks do average to zero, but their effect on the *position* does not: the random walk shows that a sum of zero-mean uncorrelated steps has a mean square that grows linearly with their number. The correct theory was given by Einstein in 1905.

1.4.1. Separation of time scales

Einstein's key insight is that there are **two very different time scales** in the problem.

1. **The microscopic collision time τ** — the interval between successive molecular impacts on the Brownian particle.
2. **The macroscopic dissipation time t_d** — the time over which the Brownian particle loses the memory of its velocity because of the viscous friction of the solvent.

The friction is described by **Stokes' law** for a sphere of radius R moving at velocity v in a fluid of viscosity η :

$$m \frac{dv}{dt} = -6\pi\eta R v \equiv -\tilde{\gamma} v \implies v(t) = v(0) e^{-t/t_d}, \quad t_d = \frac{m}{6\pi\eta R} = \frac{m}{\tilde{\gamma}}.$$

Stokes' law is a *macroscopic*, hydrodynamic law: it already contains the average effect of countless collisions. It applies only on times much longer than τ . For a particle of radius $R \simeq 1 \mu\text{m}$ with a density close to that of water, at room temperature, $t_d = \mathcal{O}(10^{-7} \text{ s})$, while $\tau = \mathcal{O}(10^{-11} \text{ s})$.

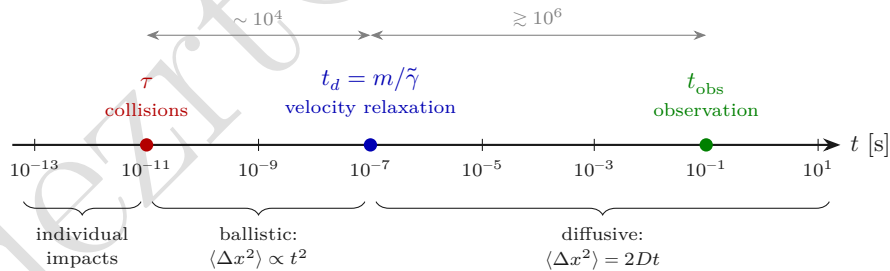


Figure: the hierarchy of time scales for a micrometre-sized colloid in water (logarithmic axis). Thousands of collisions occur during one velocity-relaxation time, and millions of relaxation times elapse between two observations. On the observation scale the velocity is completely decorrelated and the motion is diffusive.

The separation $\tau \ll t_d$ allows two complementary descriptions: on scales $\gg \tau$ the solvent acts as a continuum exerting friction (Stokes) plus a rapidly fluctuating force; on scales $\gg t_d$ the Brownian particle has forgotten its velocity and performs a random walk. Einstein treated Brownian particles as “**big molecules**”: from the thermodynamic point of view they are a dilute solute in the solvent.

1.4.2. Einstein's theory: sedimentation equilibrium

To relate diffusion to friction, Einstein considered a *non-uniform* stationary state. Take an ensemble of Brownian particles in a closed container, subject to gravity. Macroscopic heavy bodies would sit on the

bottom; Brownian particles instead form a density profile $n(z)$ that decreases with height, $\partial_z n < 0$ and $n \rightarrow 0$ as $z \rightarrow \infty$. This profile is stationary, and we can describe it in two equivalent ways.

Balance of currents (kinetics). Under the constant force $F_0 = -mg$ the particle reaches the terminal (sedimentation) velocity given by the balance with Stokes friction,

$$m \frac{dv}{dt} = F_0 - \tilde{\gamma} v \quad \Rightarrow \quad v_0 = \frac{F_0}{\tilde{\gamma}}.$$

This produces a downward **drift current** nv_0 . The density gradient produces an upward **diffusion current** $-D \partial_z n$ (Fick's law). In the stationary state the total current vanishes:

$$-D \partial_z n(z) + n(z) v_0 = 0.$$

Balance of forces (thermodynamics). A dilute solution obeys **van 't Hoff's law** for the osmotic pressure, $P = nT$ — the ideal gas law for the solute. In mechanical equilibrium the osmotic pressure gradient balances the external force per unit volume:

$$-\partial_z P + n(z) F_0 = 0 \quad \Rightarrow \quad -\partial_z n(z) + \frac{F_0}{T} n(z) = 0.$$

The two equations must describe the same profile. Writing the first as $-\partial_z n + \frac{F_0}{\tilde{\gamma} D} n = 0$ and comparing:

Theorem 1.1 (Einstein relation). *For a Brownian particle in a solvent at temperature T ,*

$$D = \frac{T}{\tilde{\gamma}} = \frac{T}{6\pi\eta R}.$$

The diffusion coefficient (a property of fluctuations) is fixed by the friction coefficient (a property of dissipation) and by the temperature.

Solving either equation gives the **barometric (Perrin) profile**

$$n(z) = n(0) \exp\left(-\frac{mgz}{T}\right),$$

the Boltzmann distribution in the gravitational potential $U = mgz$. Perrin measured these profiles for colloids, extracted $T/(mg)$, and — knowing m and T in kelvin — obtained Boltzmann's constant and hence Avogadro's number. This was decisive evidence for atoms.

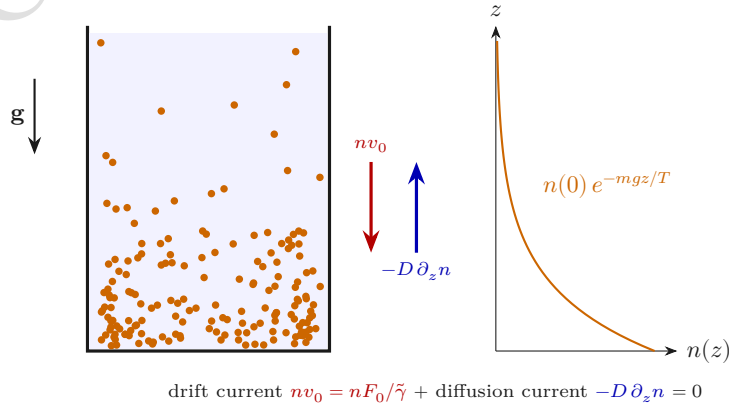


Figure: sedimentation equilibrium of Brownian particles. Gravity drives a downward drift current, the resulting density gradient drives an upward diffusive current. The stationary state, where the two cancel at

1. Random walks, Brownian motion and the Langevin equation

every height, is the exponential barometric profile. Requiring that it coincide with the thermodynamic (van 't Hoff / Boltzmann) prediction fixes $D = T/\tilde{\gamma}$.

Remark (Microscopic and macroscopic at once). Einstein's theory is a hybrid. The Brownian particles are treated *microscopically* as solute molecules (van 't Hoff's law, Boltzmann statistics) and *macroscopically* as bodies moving through a viscous continuum (Stokes' law). Classical kinetic theory would not allow an equation of state for mesoscopic objects; Einstein's point is that they are not subject to macroscopic forces but to the same thermal fluctuations as the molecules. (Strictly, m in mg is the buoyancy-corrected mass $m(1 - \rho_{\text{solvent}}/\rho_{\text{particle}})$.)

1.5. The Langevin equation

Einstein's theory describes the diffusive regime $t \gg t_d$. Paul Langevin (1908) proposed a mechanical description valid on *all* time scales $t \gg \tau$, both below and above t_d , by writing **Newton's equation with a random force**. For the first time a deterministic and a stochastic force appeared in the same equation.

The solvent acts on the Brownian particle in two ways:

- a **systematic** part, the average effect of collisions on a moving body: the Stokes friction $-\tilde{\gamma} \mathbf{v}$;
- a **fluctuating** part $\tilde{\eta}(t)$, the deviation of the instantaneous collision force from its average.

Definition 1.4 (White noise). The components $\tilde{\eta}_i(t)$, $i = 1, 2, 3$, of the stochastic force are modelled as independent, isotropic, Gaussian random processes with

$$\langle \tilde{\eta}_i(t) \rangle = 0, \quad \langle \tilde{\eta}_i(t) \tilde{\eta}_j(t') \rangle = \tilde{\Gamma} \delta_{ij} \delta(t - t').$$

The Dirac delta means that the force is **uncorrelated in time** on every scale we resolve: its true correlation time is $\mathcal{O}(\tau)$, negligible compared with t_d . Gaussianity follows from the central limit theorem, since in any resolvable time interval the force sums a huge number of nearly independent impacts. Such a process is called **(Gaussian) white noise** — its power spectrum is flat.

Definition 1.5 (Langevin equation). For each component of the velocity,

$$m \frac{dv_i}{dt} = -\tilde{\gamma} v_i(t) + \tilde{\eta}_i(t) \quad \Longleftrightarrow \quad \frac{dv_i}{dt} = -\gamma v_i(t) + \eta_i(t),$$

with $\gamma = \tilde{\gamma}/m = 1/t_d$, $\eta_i = \tilde{\eta}_i/m$, and $\langle \eta_i(t) \eta_j(t') \rangle = \Gamma \delta_{ij} \delta(t - t')$, $\Gamma = \tilde{\Gamma}/m^2$.

The equation is linear, so it can be integrated formally like any first-order ODE with a source:

$$v_i(t) = e^{-\gamma t} \left[v_i(0) + \int_0^t ds e^{\gamma s} \eta_i(s) \right].$$

Each realisation of the noise gives a different trajectory $v_i(t)$; the physics is in the **averages over realisations**.

1.5.1. Relaxation of the velocity and the fluctuation–dissipation relation

First moment. Since $\langle \eta_i \rangle = 0$,

$$\langle v_i(t) \rangle = v_i(0) e^{-\gamma t} :$$

the mean velocity decays exactly as for a deterministic particle under Stokes friction. The memory of the initial condition is lost on the time scale t_d .

Second moment. Squaring the solution, the terms linear in the noise vanish on average, and the double integral is collapsed by the delta function:

$$\begin{aligned}\langle v_i^2(t) \rangle &= e^{-2\gamma t} v_i^2(0) + e^{-2\gamma t} \int_0^t ds \int_0^t ds' e^{\gamma(s+s')} \langle \eta_i(s) \eta_i(s') \rangle \\ &= e^{-2\gamma t} v_i^2(0) + \Gamma e^{-2\gamma t} \int_0^t ds e^{2\gamma s} = v_i^2(0) e^{-2\gamma t} + \frac{\Gamma}{2\gamma} (1 - e^{-2\gamma t}).\end{aligned}$$

While the mean decays to zero, the mean square does **not**: it relaxes to the finite value

$$\lim_{t \rightarrow \infty} \langle v_i^2(t) \rangle = \frac{\Gamma}{2\gamma}.$$

Friction removes kinetic energy, noise pumps it back in, and the balance sets a stationary level of fluctuations.

So far Γ is a free parameter. The bridge to thermodynamics is to require that, at long times, the Brownian particle be in **thermal equilibrium** with the solvent at temperature T . Then equipartition gives $\frac{1}{2}m\langle v_i^2 \rangle = \frac{1}{2}T$, hence

Theorem 1.2 (Fluctuation–dissipation relation (Langevin)).

$$\Gamma = \frac{2\gamma T}{m} \quad \Longleftrightarrow \quad \tilde{\Gamma} = 2\tilde{\gamma} T.$$

The strength of the random force is proportional to the friction coefficient and to the temperature.

This is the first and simplest example of a **fluctuation–dissipation relation**: friction and noise are *not independent* — they are two faces of the same molecular collisions. A solvent that damps motion strongly must also kick strongly, otherwise the particle would cool below T (if Γ were too small) or heat above it (if too large). The relation will reappear, in a much more general form, in linear response theory.

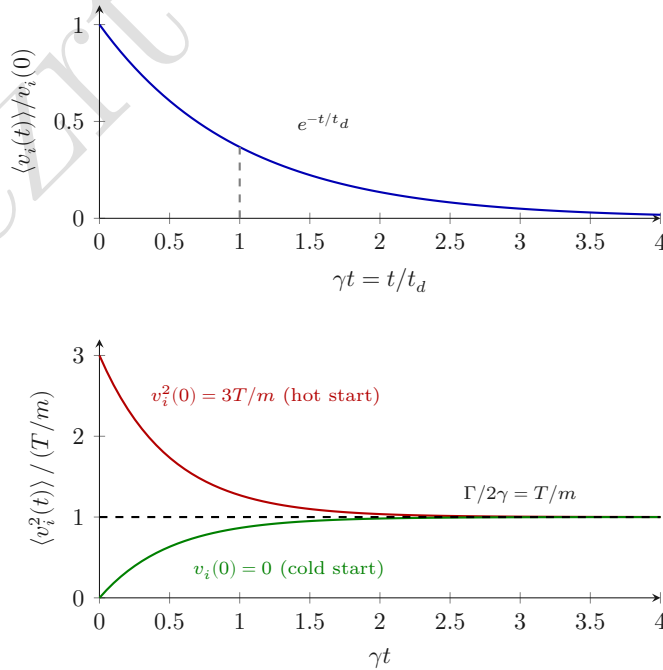


Figure: relaxation in the Langevin model. Top: the mean velocity decays exponentially on the dissipation

1. Random walks, Brownian motion and the Langevin equation

time t_d . Bottom: the mean square velocity reaches the equipartition value T/m from above or from below, on the time scale $t_d/2$. Whatever the initial condition, the particle thermalises with the bath — the fluctuation–dissipation relation is what makes the final plateau equal to T/m .

1.5.2. Ballistic and diffusive regimes

The velocity is not directly measurable (the trajectories are too erratic), but the **mean square displacement** is. Since $x_i(t) - x_i(0) = \int_0^t v_i(s) ds$,

$$\langle (x_i(t) - x_i(0))^2 \rangle = \int_0^t ds \int_0^t ds' \langle v_i(s) v_i(s') \rangle,$$

which requires the **velocity correlation function**. From the formal solution, and integrating the delta function first over the later of the two times,

$$\langle v_i(s) v_i(s') \rangle = v_i^2(0) e^{-\gamma(s+s')} + \frac{\Gamma}{2\gamma} (e^{-\gamma|s-s'|} - e^{-\gamma(s+s')}).$$

For $s = s'$ this reduces to $\langle v_i^2(s) \rangle$ above. At long times both $s, s' \gg t_d$ and only the stationary part survives:

$$\langle v_i(s) v_i(s') \rangle \rightarrow \frac{T}{m} e^{-\gamma|s-s'|}.$$

velocities at different times are correlated over a time t_d only.

Integrating, with the help of

$$\int_0^t \int_0^t ds ds' e^{-\gamma(s+s')} = \left(\frac{1 - e^{-\gamma t}}{\gamma} \right)^2, \quad \int_0^t \int_0^t ds ds' e^{-\gamma|s-s'|} = \frac{2t}{\gamma} - \frac{2}{\gamma^2} (1 - e^{-\gamma t}),$$

one finds

$$\langle (x_i(t) - x_i(0))^2 \rangle = \left(v_i^2(0) - \frac{\Gamma}{2\gamma} \right) \frac{(1 - e^{-\gamma t})^2}{\gamma^2} + \frac{\Gamma}{\gamma^2} t - \frac{\Gamma}{\gamma^3} (1 - e^{-\gamma t}).$$

Proposition 1.3 (Ballistic and diffusive regimes).

$$\langle (x_i(t) - x_i(0))^2 \rangle \simeq \begin{cases} v_i^2(0) t^2, & t \ll t_d \quad (\text{ballistic regime}), \\ \frac{\Gamma}{\gamma^2} t = 2Dt, & t \gg t_d \quad (\text{diffusive regime}). \end{cases}$$

In the diffusive regime, using $\Gamma = 2\gamma T/m$,

$$D = \frac{\Gamma}{2\gamma^2} = \frac{T}{m\gamma} = \frac{T}{\tilde{\gamma}},$$

which is **Einstein's relation**, now derived from the microscopic model plus equipartition. In d dimensions, $\lim_{t \rightarrow \infty} \langle (\mathbf{x}(t) - \mathbf{x}(0))^2 \rangle / t = 2dD$.

For $t \ll t_d$ expand $1 - e^{-\gamma t} \simeq \gamma t - \gamma^2 t^2/2$: the Γ terms cancel at order t^2 and leave $v_i^2(0)t^2$ — the particle flies freely with its initial velocity. For $t \gg t_d$, $1 - e^{-\gamma t} \simeq 1$ and the linear term dominates.

If the particle is already thermalised at $t = 0$ (average over $v_i(0)$ with $\langle v_i^2(0) \rangle = T/m = \Gamma/2\gamma$), the first term disappears and the result takes the compact **Ornstein–Furth** form

$$\langle (x_i(t) - x_i(0))^2 \rangle = 2D [t - t_d (1 - e^{-t/t_d})],$$

which is plotted below.

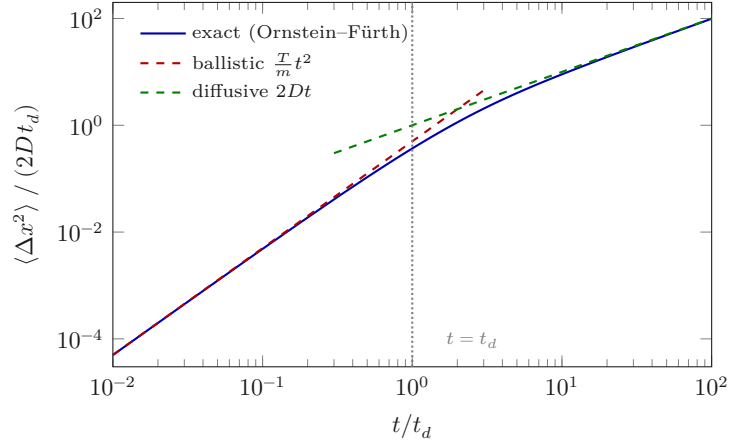


Figure: mean square displacement of a thermalised Brownian particle (log-log). Slope 2 at short times (free flight at the thermal speed $\sqrt{T/m}$), slope 1 at long times (diffusion). The crossover happens at the dissipation time t_d : the particle needs a few t_d to “forget” its velocity and start behaving as a random walker.

Remark (Why Brown only saw diffusion). For a $1\mu\text{m}$ colloid in water at room temperature, $m \simeq 4 \times 10^{-15}\text{ kg}$, the thermal speed is $\sqrt{T/m} \sim 10^{-3}\text{ m/s}$ and $t_d \sim 2 \times 10^{-7}\text{ s}$: the ballistic flights are $\sim 10^{-10}\text{ m}$ long, the size of an atom. Meanwhile $D = T/(6\pi\eta R) \simeq 2 \times 10^{-13}\text{ m}^2/\text{s}$, giving $\sqrt{2Dt} \approx 0.7\mu\text{m}$ in one second — easily visible. The ballistic regime of a Brownian particle was resolved experimentally only in 2010, with optical tweezers.

Remark (Diffusion from velocity correlations). The long-time result can be read as

$$D = \int_0^\infty \langle v_i(0) v_i(s) \rangle_{\text{eq}} ds = \int_0^\infty \frac{T}{m} e^{-\gamma s} ds = \frac{T}{m\gamma}.$$

A transport coefficient equals the time integral of an equilibrium correlation function. This is the simplest instance of a **Green–Kubo relation**, which we will meet again in linear response theory.

1.5.3. Langevin equation with a conservative force

The generalisation to a particle in an external potential $U(\mathbf{x})$, with force $\mathbf{F} = -\nabla U$, is immediate:

$$m \ddot{\mathbf{x}}(t) = -\nabla U(\mathbf{x}) - \tilde{\gamma} \dot{\mathbf{x}}(t) + \tilde{\eta}(t), \quad \langle \tilde{\eta}_i(t) \tilde{\eta}_j(t') \rangle = 2\tilde{\gamma} T \delta_{ij} \delta(t - t').$$

The noise is unchanged: it describes the solvent, which does not care about the external field. The fluctuation–dissipation relation guarantees that the stationary state is the Boltzmann distribution $\propto e^{-[\frac{1}{2}mv^2 + U(\mathbf{x})]/T}$ (we verify this in Chapter 2 for the position).

Remark (Overdamped limit). On time scales $t \gg t_d$ inertia is negligible: the velocity relaxes instantaneously to the value set by the forces. Dropping $m\ddot{\mathbf{x}}$,

$$\dot{\mathbf{x}} = \frac{\mathbf{F}(\mathbf{x})}{\tilde{\gamma}} + \frac{\tilde{\eta}(t)}{\tilde{\gamma}}, \quad \left\langle \frac{\tilde{\eta}_i(t)}{\tilde{\gamma}} \frac{\tilde{\eta}_j(t')}{\tilde{\gamma}} \right\rangle = \frac{2T}{\tilde{\gamma}} \delta_{ij} \delta(t - t') = 2D \delta_{ij} \delta(t - t').$$

The position performs a random walk with **drift velocity** $\mathbf{F}/\tilde{\gamma}$ (the sedimentation speed v_0 for gravity) and diffusion coefficient D . This *overdamped Langevin equation* is the natural partner of the Fokker–Planck

equation of Chapter 2.

1.6. The Fokker–Planck approach: the master equation

The Langevin equation gives individual stochastic trajectories, and the physics is extracted by averaging over many of them. Following Boltzmann’s spirit, one can instead get rid of trajectories from the start and describe the **probability density** $p(\mathbf{x}, t)$, such that $p(\mathbf{x}, t) d^d x$ is the probability of finding the Brownian particle in the volume $d^d x$ around $\mathbf{x}$ at time t .

We coarse-grain time on a scale $\gg t_d$ (so any reference to individual collisions, and to the velocity, is lost) and describe the motion as a sequence of random **jumps** in space, characterised by transition rates.

Definition 1.6 (Transition rate). $R(\mathbf{x}, \mathbf{x}')$ is the **rate of transitions from $\mathbf{x}'$ to $\mathbf{x}$** : $R(\mathbf{x}, \mathbf{x}') d^d x \Delta t$ is the probability that a particle at $\mathbf{x}'$ jumps into the volume $d^d x$ around $\mathbf{x}$ during a short interval Δt . It is a probability per unit time *and* per unit volume. The process is **Markovian**: the rate depends only on the present position, not on the past history.

(Livi–Politi call this $W(\mathbf{x}, \mathbf{x}')$ in §1.4.2. We use R , the same symbol as for the rates R_{ik} of the continuous-time master equation of §1.6, with the same convention: *first index = arrival, second index = departure*.)

The time evolution of p is a **balance equation**: the probability at $\mathbf{x}$ grows because of jumps *into* $\mathbf{x}$ from everywhere else (gain) and decreases because of jumps *out of* $\mathbf{x}$ towards everywhere else (loss).

Definition 1.7 (Master equation).

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} = \int d^d x' \left[\underbrace{R(\mathbf{x}, \mathbf{x}') p(\mathbf{x}', t)}_{\text{gain: } \mathbf{x}' \rightarrow \mathbf{x}} - \underbrace{R(\mathbf{x}', \mathbf{x}) p(\mathbf{x}, t)}_{\text{loss: } \mathbf{x} \rightarrow \mathbf{x}'} \right].$$

For a discrete set of states i the integral becomes a sum,

$$\frac{dp_i(t)}{dt} = \sum_{k \neq i} [R_{ik} p_k(t) - R_{ki} p_i(t)].$$

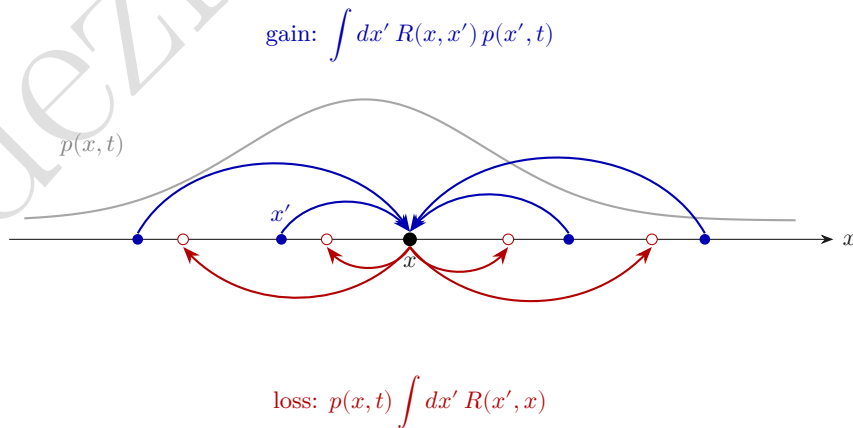


Figure: the master equation as a balance of probability fluxes at the point x . Blue arrows bring probability in from every x' , weighted by how much probability sits at x' ; red arrows carry probability out of x , weighted by $p(x, t)$. Only the net flux changes $p(x, t)$.

Two properties follow directly from the structure of the equation.

- **Conservation of probability.** Integrating over $\mathbf{x}$ and exchanging the names of the dummy variables $\mathbf{x} \leftrightarrow \mathbf{x}'$ in the loss term, the two terms cancel: $\frac{d}{dt} \int d^d x p(\mathbf{x}, t) = 0$.
- **Linearity.** The master equation is linear in p , even when the underlying Langevin dynamics is nonlinear. The price is that it is an integro-differential equation, generally hard to solve.

Example 1.2 (*Hopping on a lattice gives diffusion*)

Let a particle hop between the sites $x_n = na$ of a one-dimensional lattice, jumping to each nearest neighbour with rate r . The master equation reads

$$\frac{dp_n}{dt} = r p_{n-1} + r p_{n+1} - 2r p_n.$$

If $p_n(t)$ varies slowly on the lattice scale, set $p_n(t) = a p(x_n, t)$ and Taylor-expand $p(x \pm a) = p \pm a \partial_x p + \frac{a^2}{2} \partial_x^2 p + \dots$. The first-order terms cancel by symmetry and

$$\frac{\partial p}{\partial t} = ra^2 \frac{\partial^2 p}{\partial x^2} \equiv D \frac{\partial^2 p}{\partial x^2}, \quad D = ra^2.$$

Small jumps plus a smooth density turn the master equation into a *differential* equation — the diffusion equation. Chapter 2 does this expansion in general, keeping the possibility of asymmetric (drifting) and position-dependent jumps: the result is the Fokker–Planck equation.

Remark (Three levels of description). We now have three descriptions of the same physics, from fine to coarse:

1. **Microscopic mechanics:** Newton's equations for the particle *and* all solvent molecules (unsolvable, and useless anyway).
2. **Langevin equation:** the particle's own degrees of freedom, with the solvent replaced by friction plus noise (valid for $t \gg \tau$). Gives stochastic trajectories.
3. **Master / Fokker–Planck equation:** a deterministic, linear equation for the probability density (valid on coarse-grained times). Gives distributions directly.

The fluctuation–dissipation relation and the Einstein relation are the glue that makes the reduced descriptions consistent with equilibrium thermodynamics.

1.7. Exercises

Exercise 1.1

(Mean free path of air) Treat air as an ideal gas of molecules of effective radius $r \simeq 1.8 \times 10^{-10}$ m at $T = 293$ K and $P = 1$ atm. Estimate n , the cross section σ , the mean free path $\lambda = 1/(\sqrt{2}n\sigma)$, and the collision time $\tau = \lambda/\langle v \rangle$ (take $m = 29$ u). Compare λ with $n^{-1/3}$ and r .

Solution

$n = P/(k_B T) \simeq 1.01 \times 10^5 / (1.38 \times 10^{-23} \cdot 293) \simeq 2.5 \times 10^{25} \text{ m}^{-3}$. $\sigma = 4\pi r^2 \simeq 4.1 \times 10^{-19} \text{ m}^2$. $\lambda \simeq 1/(1.414 \cdot 2.5 \times 10^{25} \cdot 4.1 \times 10^{-19}) \simeq 7 \times 10^{-8} \text{ m}$. $\langle v \rangle = \sqrt{8k_B T/(\pi m)} \simeq 460 \text{ m/s}$, so $\tau \simeq 1.5 \times 10^{-10} \text{ s}$. The hierarchy is $r \sim 10^{-10} \text{ m} \ll n^{-1/3} \simeq 3.4 \times 10^{-9} \text{ m} \ll \lambda \sim 10^{-7} \text{ m}$.

Exercise 1.2

(Persistent random walk) In a one-dimensional walk with steps $\pm\ell$, suppose each step repeats the direction of the previous one with probability q and reverses it with probability $1 - q$, so that $\langle x_i x_j \rangle = \ell^2 c^{|i-j|}$ with $c = 2q - 1$. Show that for large N

$$\langle X^2 \rangle \simeq \ell^2 N \frac{1+c}{1-c}.$$

Interpret the result in terms of an effective step length, and discuss the limits $q \rightarrow 1$ and $q \rightarrow 0$.

Solution

$\langle X^2 \rangle = \ell^2 \sum_{i,j} c^{|i-j|} = \ell^2 [N + 2 \sum_{k=1}^{N-1} (N-k)c^k] \simeq \ell^2 N [1 + 2c/(1-c)] = \ell^2 N (1+c)/(1-c)$ for $N \gg 1/(1-c)$. The walk is still diffusive, but correlations renormalise D : it behaves as an uncorrelated walk whose effective step is longer by $\sqrt{(1+c)/(1-c)}$. For $q \rightarrow 1$ ($c \rightarrow 1$) the walker almost never turns and D diverges (ballistic limit); for $q \rightarrow 0$ ($c \rightarrow -1$) it oscillates back and forth and $D \rightarrow 0$. This is the discrete analogue of the finite velocity correlation time t_d in the Langevin model.

Exercise 1.3

(Velocity autocorrelation in equilibrium) Starting from the formal solution of the Langevin equation and assuming that $v_i(0)$ is drawn from the equilibrium distribution, independent of the noise, show that $\langle v_i(s) v_i(s') \rangle = (T/m) e^{-\gamma|s-s'|}$ for all $s, s' \geq 0$, i.e. that the process is stationary. Then derive the Ornstein–Förth formula for the mean square displacement.

Exercise 1.4

(Equipartition fixes the noise) A Brownian particle is trapped in a harmonic potential $U = \frac{1}{2}\kappa x^2$ in the overdamped regime: $\tilde{\gamma}\dot{x} = -\kappa x + \tilde{\eta}(t)$, $\langle \tilde{\eta}(t)\tilde{\eta}(t') \rangle = \tilde{\Gamma}\delta(t-t')$. Solve for $x(t)$, compute $\lim_{t \rightarrow \infty} \langle x^2(t) \rangle$, and show that equipartition $\frac{1}{2}\kappa \langle x^2 \rangle = \frac{1}{2}T$ again requires $\tilde{\Gamma} = 2\tilde{\gamma}T$. What is the relaxation time of $\langle x(t) \rangle$?

Solution

$x(t) = x_0 e^{-t/t_\kappa} + \tilde{\gamma}^{-1} \int_0^t ds e^{-(t-s)/t_\kappa} \tilde{\eta}(s)$ with $t_\kappa = \tilde{\gamma}/\kappa$. Then $\langle x^2 \rangle \rightarrow \tilde{\Gamma} t_\kappa / (2\tilde{\gamma}^2) = \tilde{\Gamma} / (2\tilde{\gamma}\kappa)$. Imposing $\kappa \langle x^2 \rangle = T$ gives $\tilde{\Gamma} = 2\tilde{\gamma}T$, the same relation as for the free particle: the noise is a property of the bath, not of the potential. $\langle x(t) \rangle$ relaxes on $t_\kappa = \tilde{\gamma}/\kappa$.

Exercise 1.5

(Perrin's measurement) Colloidal spheres of radius $R = 0.21 \mu\text{m}$ and density $\rho_p = 1.19 \text{ g/cm}^3$ are suspended in water ($\rho = 1.00 \text{ g/cm}^3$) at $T = 293 \text{ K}$. Compute the height $\ell_g = T/(m_{\text{eff}}g)$ over which the density decreases by a factor e . If one measured ℓ_g in this way without knowing k_B , how would one obtain Avogadro's number from the gas constant $\mathcal{R}$?

Exercise 1.6

(Master equation conserves probability) Prove that the master equation preserves normalisation and positivity: if $p(\mathbf{x}, 0) \geq 0$ and $\int p(\mathbf{x}, 0) d^d x = 1$, the same holds at all times. (Hint for positivity: look at a point where p first touches zero.)

1.8. Sources

Lecture 1 of the course: random walk as a basic model of diffusion and kinetic estimates of mean free path and collision time; Brownian motion and the separation between the microscopic collision time and the macroscopic dissipation time; the Langevin equation, the Einstein relation as a first fluctuation–dissipation relation, ballistic and diffusive regimes, conservative forces; the master equation for the probability density with transition rates $R(\mathbf{x}, \mathbf{x}')$. Book: R. Livi and P. Politi, *Nonequilibrium Statistical Physics: A Modern Perspective*, Cambridge (2017), §1.1, §1.2.1–1.2.2, §1.3 (transport coefficients, for context), §1.4 (introduction), §1.4.1 and the first part of §1.4.2; the discrete master equation is from §1.5.4 and §1.6 (introduction). Original papers: A. Einstein, *Ann. Phys.* **17**, 549 (1905); P. Langevin, *C. R. Acad. Sci.* **146**, 530 (1908); J. Perrin, *Les Atomes* (1913).

2. The Fokker–Planck equation

Chapter 1 ended with the master equation, a balance law for the probability density $p(\mathbf{x}, t)$ driven by transition rates $R(\mathbf{x}, \mathbf{x}')$. It is exact but an integro-differential equation, and its rates are rarely known in detail. This chapter shows that when jumps are **small** — the generic situation for a Brownian particle observed on coarse-grained times — the master equation reduces to a second-order partial differential equation, the **Fokker–Planck equation**, which depends on the rates only through two numbers at each point: a **drift velocity** and a **diffusion coefficient**. We solve it in the simplest case (Gaussian spreading), recast it as a **continuity equation** for probability with an explicit **probability current**, and use the current to make precise the difference between a *stationary* state and an *equilibrium* state. Requiring equilibrium in a conservative force field gives back the Boltzmann distribution and, once more, the Einstein relation. Finally we derive the same equation from a different starting point, the **Chapman–Kolmogorov equation** for the transition probability, which comes in a *forward* and a *backward* version.

2.1. From the master equation to the Fokker–Planck equation

2.1.1. Jump variables

Start from the master equation of Chapter 1, in one dimension for clarity (the d -dimensional case is identical with vector indices):

$$\frac{\partial p(x, t)}{\partial t} = \int dx' [R(x, x') p(x', t) - R(x', x) p(x, t)],$$

where $R(x, x')$ is the rate of transitions *from* x' *to* x . It is convenient to describe a transition by its **departure point** and its **jump** χ (arrival minus departure), and to write

$$R(y; \chi) \equiv R(y + \chi, y) = \text{rate of a jump of size } \chi \text{ starting from } y.$$

- In the **gain** term the particle departs from $x' = x - \chi$ and arrives at x : $R(x, x') = R(x - \chi; \chi)$, $dx' = d\chi$ (up to an irrelevant sign absorbed by flipping the integration limits).
- In the **loss** term the particle departs from x and arrives at $x' = x + \chi$: $R(x', x) = R(x; \chi)$.

Hence

$$\frac{\partial p(x, t)}{\partial t} = \int d\chi [R(x - \chi; \chi) p(x - \chi, t) - R(x; \chi) p(x, t)].$$

Both terms involve the *same* function of the departure point, $f_\chi(y) \equiv R(y; \chi) p(y, t)$, evaluated at $y = x - \chi$ and at $y = x$.

2.1.2. The small-jump expansion

Brownian particles do not make large jumps: $R(y; \chi)$ is appreciably different from zero only for $|\chi|$ smaller than some microscopic length ℓ . On the other hand, p and the *dependence of the rates on the departure point* vary on a macroscopic scale $L \gg \ell$. We can therefore Taylor-expand $f_\chi(x - \chi)$ around $f_\chi(x)$ — in its **first argument only**:

$$R(x - \chi; \chi) p(x - \chi, t) = R(x; \chi) p(x, t) - \chi \frac{\partial}{\partial x} [R(x; \chi) p(x, t)] + \frac{\chi^2}{2} \frac{\partial^2}{\partial x^2} [R(x; \chi) p(x, t)] - \dots$$

2. The Fokker–Planck equation

The zeroth-order term cancels exactly against the loss term. Exchanging the χ -integral with the x -derivatives,

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x} \left[\alpha(x) p(x, t) \right] + \frac{1}{2} \frac{\partial^2}{\partial x^2} \left[\beta(x) p(x, t) \right] + \dots$$

with the first two **jump moments**

$$\alpha(x) = \int d\chi \chi R(x; \chi), \quad \beta(x) = \int d\chi \chi^2 R(x; \chi).$$

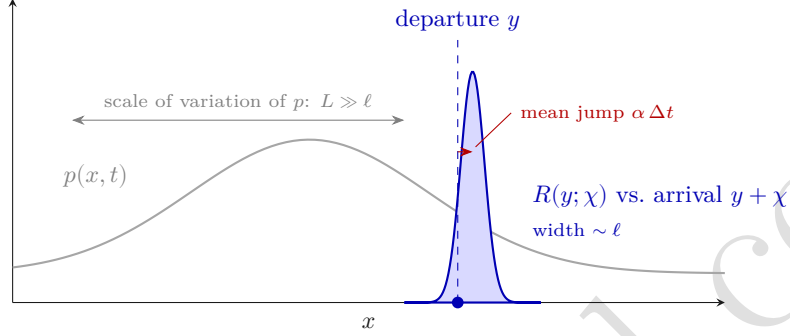


Figure: the assumption behind the Fokker–Planck equation. From any departure point y the jump kernel $R(y; \chi)$ is sharply peaked (width ℓ), while the probability density and the y -dependence of the kernel vary on a much larger scale L . The kernel may be asymmetric (its mean is the drift); only its first two moments survive in the equation.

Theorem 2.1 (Kramers–Moyal expansion). *Keeping all orders, the master equation is equivalent to*

$$\frac{\partial p(x, t)}{\partial t} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \frac{\partial^n}{\partial x^n} \left[M_n(x) p(x, t) \right], \quad M_n(x) = \int d\chi \chi^n R(x; \chi).$$

Why may we stop at $n = 2$? If jumps of size $\sim \ell$ occur at a rate $\sim 1/\tau_j$, then $M_n \sim \ell^n/\tau_j$ and each derivative acting on p brings a factor $1/L$, so the n -th term is of relative order $(\ell/L)^n$. More sharply, in the **diffusive scaling limit** $\ell \rightarrow 0$, $\tau_j \rightarrow 0$ with ℓ^2/τ_j fixed (exactly how the random walk becomes Brownian motion), M_1 and M_2 remain finite while $M_{n \geq 3} \sim \ell^{n-2} \cdot \ell^2/\tau_j \rightarrow 0$.

Remark (Truncate at two, or not at all). Pawula’s theorem states that a truncation of the Kramers–Moyal series at any finite order $n > 2$ is inconsistent with the positivity of p . The expansion must stop at second order (Fokker–Planck) or be kept in full (the master equation). Note also that we never needed $R(y; \chi)$ to be a smooth function of the jump χ : it can be arbitrarily sharp. Smoothness is required only in the departure point y .

Definition 2.1 (Fokker–Planck equation). In d dimensions, with $\mathbf{x} = (x_1, \dots, x_d)$,

$$\boxed{\frac{\partial p(\mathbf{x}, t)}{\partial t} = -\sum_i \frac{\partial}{\partial x_i} \left[\alpha_i(\mathbf{x}) p(\mathbf{x}, t) \right] + \frac{1}{2} \sum_{i,j} \frac{\partial^2}{\partial x_i \partial x_j} \left[\beta_{ij}(\mathbf{x}) p(\mathbf{x}, t) \right]}$$

with drift vector and diffusion matrix

$$\alpha_i(\mathbf{x}) = \int d^d \chi \chi_i R(\mathbf{x}; \chi), \quad \beta_{ij}(\mathbf{x}) = \int d^d \chi \chi_i \chi_j R(\mathbf{x}; \chi).$$

β_{ij} is symmetric and positive semi-definite.

Although we had the Brownian particle in mind, nothing in the derivation refers to it: $\mathbf{x}$ can be any set of variables describing the state of a system (a velocity, a chemical concentration, a price, a population). That is why the Fokker–Planck equation is ubiquitous.

2.2. Drift velocity and diffusion coefficient

The jump moments have a transparent meaning. In a short time Δt , a particle at $\mathbf{x}$ jumps by χ with probability $R(\mathbf{x}; \chi) d^d \chi \Delta t$ (and does not move with the remaining probability). Therefore its displacement $\Delta \mathbf{x}$ satisfies

$$\langle \Delta x_i \rangle_{\mathbf{x}} = \alpha_i(\mathbf{x}) \Delta t, \quad \langle \Delta x_i \Delta x_j \rangle_{\mathbf{x}} = \beta_{ij}(\mathbf{x}) \Delta t,$$

where $\langle \cdot \rangle_{\mathbf{x}}$ is the average over jumps starting from $\mathbf{x}$.

- $\alpha = \langle \Delta \mathbf{x} \rangle / \Delta t$ is the **drift velocity**: the mean velocity of the particle. Without external forces it vanishes by symmetry. With a constant external force $\mathbf{F}_0$ (gravity) it is the **sedimentation velocity** $\mathbf{v}_0 = \mathbf{F}_0 / \tilde{\gamma}$, set by the balance of the force with Stokes friction. In general $\alpha = \mathbf{F}(\mathbf{x}) / \tilde{\gamma}$.
- $\beta_{ij} = \langle \Delta x_i \Delta x_j \rangle / \Delta t$ is the rate of growth of the mean square displacement. In a homogeneous, isotropic medium it is diagonal and constant:

$$\beta_{ij} = 2D \delta_{ij},$$

consistently with $\langle \Delta x_i^2 \rangle = 2D \Delta t$ for each component.

Note the different scaling of the two moments with Δt : the *mean* displacement is $\propto \Delta t$ while the *typical* displacement is $\propto \sqrt{\Delta t}$. Over short times fluctuations dominate the drift; over long times the drift wins.

Proposition 2.2 (Fokker–Planck equation for the Brownian particle). *With $\alpha = \mathbf{v}_0$ and $\beta_{ij} = 2D\delta_{ij}$ constant,*

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} = -\mathbf{v}_0 \cdot \nabla p(\mathbf{x}, t) + D \nabla^2 p(\mathbf{x}, t).$$

*Without external forces it reduces to the **diffusion equation** $\partial_t p = D \nabla^2 p$.*

Remark (Link with the overdamped Langevin equation). The overdamped Langevin equation of Chapter 1, $\dot{\mathbf{x}} = \mathbf{F}/\tilde{\gamma} + \xi(t)$ with $\langle \xi_i(t) \xi_j(t') \rangle = 2D\delta_{ij}\delta(t - t')$, gives in a short time $\langle \Delta \mathbf{x} \rangle = (\mathbf{F}/\tilde{\gamma})\Delta t$ and $\langle \Delta x_i \Delta x_j \rangle = 2D\delta_{ij}\Delta t + \mathcal{O}(\Delta t^2)$. Its trajectories are therefore distributed according to the Fokker–Planck equation with $\alpha = \mathbf{F}/\tilde{\gamma}$ and $\beta_{ij} = 2D\delta_{ij}$. Langevin (trajectories) and Fokker–Planck (probabilities) are two faces of the same stochastic process.

Example 2.1 (Drift and diffusion from a biased lattice walk)

A particle on a lattice of spacing a jumps right with rate r_+ and left with rate r_- : $R(x; \chi) = r_+ \delta(\chi - a) + r_- \delta(\chi + a)$. Then

$$\alpha = (r_+ - r_-)a, \quad \beta = (r_+ + r_-)a^2 \Rightarrow D = \frac{1}{2}(r_+ + r_-)a^2.$$

The asymmetry of the jumps produces the drift, their total rate produces the diffusion. For $r_+ = r_- = r$ we recover $D = ra^2$ of the example in Chapter 1.

2.2.1. Moments of the distribution

Without solving the Fokker–Planck equation one can derive closed equations for the moments, by multiplying by x or x^2 and integrating by parts (boundary terms vanish because $p \rightarrow 0$ fast at infinity). In one dimension:

$$\frac{d\langle x \rangle}{dt} = \langle \alpha(x) \rangle, \quad \frac{d\langle x^2 \rangle}{dt} = 2\langle x \alpha(x) \rangle + \langle \beta(x) \rangle.$$

For constant $\alpha = v_0$ and $\beta = 2D$: $\langle x \rangle = x_0 + v_0 t$ and $\frac{d}{dt}(\langle x^2 \rangle - \langle x \rangle^2) = 2D$, i.e.

$$\langle x^2 \rangle - \langle x \rangle^2 = 2Dt.$$

The centre of the distribution moves ballistically with the drift; its width grows diffusively.

2.3. Gaussian solution of the diffusion equation

Solve

$$\frac{\partial p}{\partial t} = -v_0 \frac{\partial p}{\partial x} + D \frac{\partial^2 p}{\partial x^2}, \quad p(x, 0) = \delta(x - x_0)$$

on the whole line. The equation is linear with constant coefficients, so it is diagonalised by the **Fourier transform** $\hat{p}(k, t) = \int dx e^{-ikx} p(x, t)$, under which $\partial_x \rightarrow ik$:

$$\frac{\partial \hat{p}}{\partial t} = (-ikv_0 - Dk^2)\hat{p}, \quad \hat{p}(k, 0) = e^{-ikx_0} \implies \hat{p}(k, t) = e^{-ik(x_0 + v_0 t)} e^{-Dk^2 t}.$$

Each Fourier mode is translated by the drift and damped at a rate Dk^2 : short wavelengths (sharp features) are erased fastest. Transforming back, a Gaussian integral (complete the square in k) gives:

Proposition 2.3 (Propagator of drift-diffusion).

$$p(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left[-\frac{(x - x_0 - v_0 t)^2}{4Dt}\right],$$

a Gaussian with mean $x_0 + v_0 t$ and variance $2Dt$. In d dimensions the solution factorises over components:

$$p(\mathbf{x}, t) = (4\pi Dt)^{-d/2} \exp\left[-\frac{|\mathbf{x} - \mathbf{x}_0 - \mathbf{v}_0 t|^2}{4Dt}\right], \quad \langle |\mathbf{x} - \langle \mathbf{x} \rangle|^2 \rangle = 2dDt.$$

The variance $2Dt$ coincides with the long-time mean square displacement of the Langevin equation: the Fokker–Planck description reproduces the **diffusive** regime (but not the ballistic one, which lives below the coarse-graining time t_d).

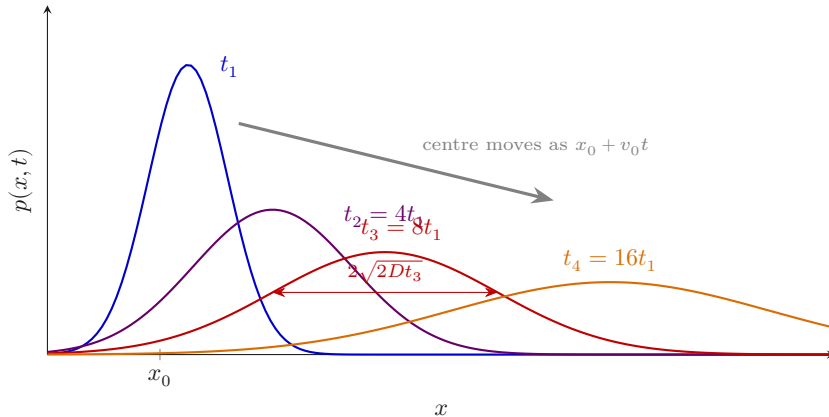


Figure: Gaussian solution of the drift-diffusion equation starting from a point at x_0 . The centre is transported at the drift velocity ($\propto t$) while the width grows only as $\sqrt{t}$; the area stays equal to one. Peak height decays as $t^{-1/2}$.

Remark (Self-similarity and the Green’s function). Without drift, $p(x, t) = (Dt)^{-1/2} \phi((x - x_0)/\sqrt{Dt})$ with $\phi(u) = e^{-u^2/4}/\sqrt{4\pi}$: the profile at all times is the same curve, rescaled. The natural variable is $x/\sqrt{t}$ — the hallmark of diffusion. Moreover, by linearity the solution for an arbitrary initial condition is the convolution $p(x, t) = \int dx_0 G(x - x_0, t) p(x_0, 0)$ with the Gaussian G above, the **Green’s function** (or **propagator**) of the diffusion equation.

2.4. The Fokker–Planck equation as a continuity equation

The Fokker–Planck equation is a total divergence. Defining the **probability current**

$$J_i(\mathbf{x}, t) = \alpha_i(\mathbf{x}) p(\mathbf{x}, t) - \frac{1}{2} \sum_j \frac{\partial}{\partial x_j} [\beta_{ij}(\mathbf{x}) p(\mathbf{x}, t)],$$

it takes the form of a **local conservation law**:

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} + \nabla \cdot \mathbf{J}(\mathbf{x}, t) = 0.$$

Probability, like mass or charge, cannot appear or disappear; it can only flow. For a Brownian particle in a force field ($\alpha = \mathbf{F}/\gamma$, $\beta_{ij} = 2D\delta_{ij}$),

$$\mathbf{J} = \underbrace{\frac{\mathbf{F}}{\gamma} p}_{\text{drift current}} - \underbrace{D \nabla p}_{\text{diffusion current (Fick)}},$$

which is exactly the decomposition used by Einstein for sedimentation (Chapter 1).

Remark (Position-dependent diffusion). If β_{ij} depends on $\mathbf{x}$, the current can be rewritten as $J_i = (\alpha_i - \frac{1}{2} \sum_j \partial_j \beta_{ij}) p - \frac{1}{2} \sum_j \beta_{ij} \partial_j p$: a gradient of the diffusion coefficient acts as an extra (sometimes called “spurious”) drift. Particles tend to accumulate where they diffuse *less*.

2.4.1. Integral form and boundary conditions

Integrating over a volume V and using Gauss’ theorem,

$$\frac{d}{dt} \int_V d^d x p(\mathbf{x}, t) = - \oint_{\partial V} \mathbf{J} \cdot d\mathbf{S} :$$

the probability inside V changes only through the flux across its boundary. In one dimension, for the interval $I = [x_1, x_2]$,

$$\frac{d}{dt} \mathcal{P}_I(t) = J(x_1, t) - J(x_2, t), \quad \mathcal{P}_I(t) = \int_{x_1}^{x_2} dx p(x, t),$$

with $J > 0$ meaning a flow towards increasing x . This makes it straightforward to formulate boundary conditions on a finite domain.

2. The Fokker–Planck equation

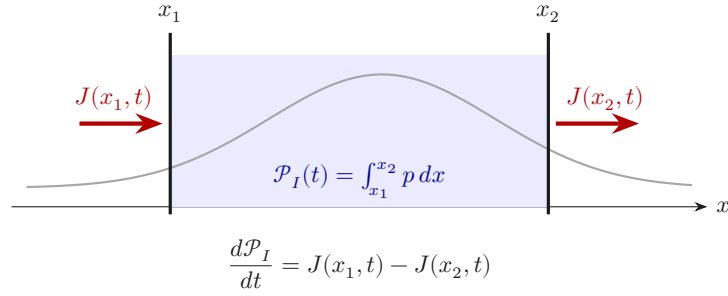


Figure: probability balance in an interval. What enters through x_1 minus what leaves through x_2 is what accumulates inside.

Definition 2.2 (Boundary conditions on an interval).

- **Reflecting** barriers: no flux crosses the boundaries, $J(x_1, t) = J(x_2, t) = 0$. The probability in I is conserved.
- **Absorbing** barriers: a walker that reaches the boundary is removed and never comes back, $p(x_1, t) = p(x_2, t) = 0$. $\mathcal{P}_I(t)$ decreases in time (it is the *survival probability*).
- **Periodic** boundary conditions: $p(x_1, t) = p(x_2, t)$ and $J(x_1, t) = J(x_2, t)$. The interval is a ring; probability is conserved because what leaves on one side re-enters on the other.

2.5. Stationary states versus equilibrium

Definition 2.3 (Stationary state and equilibrium state). A **stationary** solution $p^*(\mathbf{x})$ of the Fokker–Planck equation is time-independent, $\partial_t p^* = 0$, which by the continuity equation means

$$\nabla \cdot \mathbf{J}^*(\mathbf{x}) = 0.$$

A stationary state is an **equilibrium** state if, in addition, the current vanishes **everywhere**:

$$\mathbf{J}^*(\mathbf{x}) = 0 \quad \forall \mathbf{x}.$$

A stationary state with $\mathbf{J}^* \neq 0$ is a **nonequilibrium steady state** (NESS).

Physically, a system in thermodynamic equilibrium cannot sustain any macroscopic current — of matter, charge, heat, or probability. A stationary state only requires that currents be *divergence-free*: whatever flows into a region flows out of it, so densities do not change, but there can be steady flows. A river in steady state has a time-independent water level and a non-zero current; a lake in equilibrium has neither.

The same distinction exists for discrete states, where it takes a form that will be central to the next part of the course. For the master equation $\dot{p}_i = \sum_{k \neq i} (R_{ik} p_k - R_{ki} p_i)$,

- **stationarity** (*global balance*) requires $\sum_{k \neq i} (R_{ik} p_k^* - R_{ki} p_i^*) = 0$ for every i : the total in-flow into each state equals its total out-flow;
- **equilibrium** (*detailed balance*) requires $R_{ik} p_k^* = R_{ki} p_i^*$ for every pair (i, k) : the net current between each pair of states vanishes.

$\mathbf{J}^* = 0$ is the continuum version of detailed balance.

Proposition 2.4 (In one dimension, on the line, stationary means equilibrium). *In $d = 1$, $\partial_x J^* = 0$ implies $\mathcal{J}^*(x) = \text{const}$. If the current vanishes somewhere — at infinity with natural boundary conditions, or at a reflecting wall — it vanishes everywhere. A NESS in one dimension needs periodic*

boundary conditions, sources and sinks, or boundaries held at different densities. In $d \geq 2$ a divergence-free current can circulate (e.g. $\mathbf{J}^* = \nabla \times \mathbf{A}$) even in an infinite system.

Example 2.2 (NESS on a ring: a constant force with periodic boundaries)

Consider an overdamped particle on a ring of circumference L , pushed by a constant tangential force F (think of a colloid dragged around a circular optical trap). Then $\alpha = F/\tilde{\gamma}$, $\beta = 2D$, with periodic boundary conditions. The uniform density

$$p^*(x) = \frac{1}{L}$$

is stationary (all derivatives vanish), and the current is

$$J^* = \frac{F}{\tilde{\gamma}} \frac{1}{L} \neq 0.$$

The distribution looks like an equilibrium one, yet probability flows steadily around the ring. The force is not conservative on the ring: the would-be potential $U = -Fx$ is not single-valued (it changes by $-FL$ after one turn), so no Boltzmann distribution exists. The system is a NESS, sustained by the work that F continuously does and dissipates into the bath as heat.

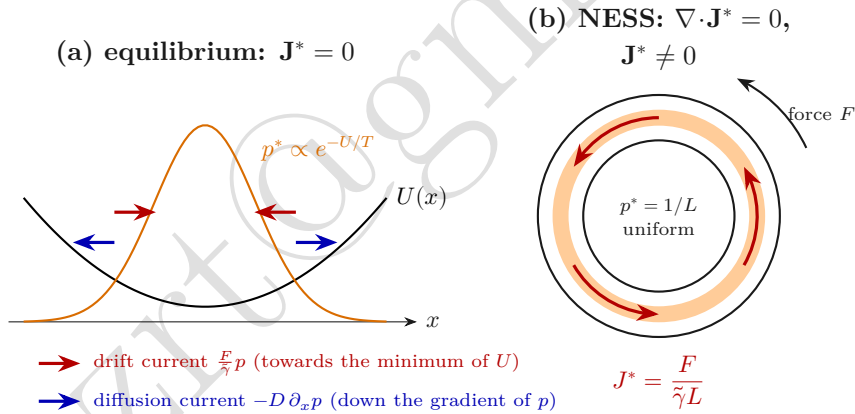


Figure: (a) In a confining potential the drift current (towards the minimum) and the diffusion current (away from the peak of p^*) cancel at every point: equilibrium, $\mathbf{J}^* = 0$. (b) On a ring with a non-conservative force the density is uniform and time-independent, but a constant current circulates: a nonequilibrium steady state.

Example 2.3 (Stationary state with a source and absorbing barriers)

Free diffusion ($\alpha = 0$, $\beta = 2D$) on $I = [x_1, x_2]$ with absorbing ends. Without sources the only stationary solution is $p^* = 0$: sooner or later every walker is absorbed. Now inject walkers at a constant rate F at $x_0 \in I$:

$$\partial_t p + \partial_x J = F \delta(x - x_0).$$

In the stationary state, away from x_0 , $\partial_x^2 p^* = 0$, so p^* is piecewise linear. Absorption gives $p^*(x_1) = p^*(x_2) = 0$, continuity at x_0 gives a tent, and integrating the equation across x_0 gives the jump in the

derivative, $-D[p^{*'}(x_0^+) - p^{*'}(x_0^-)] = F$. With $L_- = x_0 - x_1$, $L_+ = x_2 - x_0$ and $L = L_- + L_+$:

$$p^*(x) = \begin{cases} \frac{FL_+}{DL}(x - x_1), & x < x_0, \\ \frac{FL_-}{DL}(x_2 - x), & x > x_0, \end{cases} \quad J^*(x) = \begin{cases} -F L_+/L, & x < x_0, \\ +F L_-/L, & x > x_0. \end{cases}$$

The state is stationary but carries currents towards both sinks: it is a NESS maintained by the source. Two by-products:

- the probability that a walker injected at x_0 is absorbed at x_1 is $|J_-|/F = L_+/L = (x_2 - x_0)/(x_2 - x_1)$ — the closer barrier is more likely;
- the mean time spent in I is (number of walkers in I)/(out-flux) $= \frac{1}{F} \int_I p^* dx = \frac{(x_0 - x_1)(x_2 - x_0)}{2D}$.

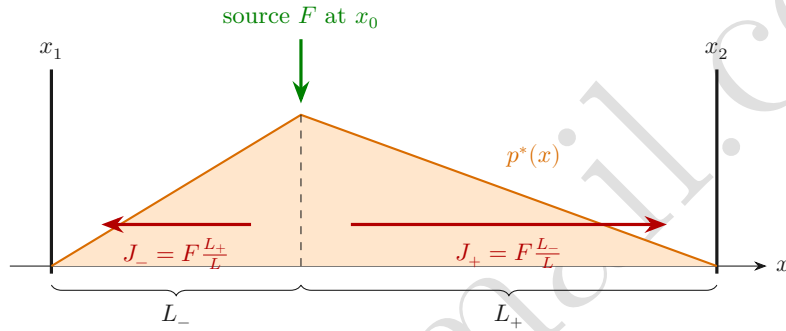


Figure: diffusion between two absorbing walls with continuous injection at x_0 . The stationary density is a tent; the slope on each side is proportional to the current flowing to that wall. More walkers leave through the nearer wall ($J_- > J_+$ when $L_- < L_+$).

2.6. Equilibrium in a conservative force field

Let the particle feel a conservative force $\mathbf{F}(\mathbf{x}) = -\nabla U(\mathbf{x})$, with $U \rightarrow \infty$ as $|\mathbf{x}| \rightarrow \infty$ (confining). The Fokker–Planck equation is

$$\frac{\partial p}{\partial t} = -\nabla \cdot \mathbf{J}, \quad \mathbf{J} = \frac{\mathbf{F}}{\tilde{\gamma}} p - D \nabla p.$$

At equilibrium the current must vanish, $\mathbf{J}^* = 0$:

$$-\frac{\nabla U}{\tilde{\gamma}} p^* = D \nabla p^* \iff \nabla \ln p^*(\mathbf{x}) = -\frac{\nabla U(\mathbf{x})}{\tilde{\gamma} D},$$

which integrates to

$$p^*(\mathbf{x}) = \frac{1}{Z} \exp\left(-\frac{U(\mathbf{x})}{\tilde{\gamma} D}\right), \quad Z = \int d^d x e^{-U/(\tilde{\gamma} D)}.$$

Thermodynamics, on the other hand, demands that a system in contact with a bath at temperature T be distributed according to **Boltzmann**, $p_{\text{eq}} \propto e^{-U/T}$. The two coincide if and only if

Theorem 2.5 (Einstein relation from the Boltzmann distribution).

$$\tilde{\gamma} D = T \iff D = \frac{T}{\tilde{\gamma}}.$$

Requiring that the zero-current stationary state of the Fokker–Planck equation be the thermodynamic equilibrium state fixes the ratio of diffusion and mobility $1/\tilde{\gamma}$.

This is the third route to the same relation: Einstein’s balance of osmotic pressure (Chapter 1), equipartition in the Langevin equation (Chapter 1), and now the Boltzmann distribution in the Fokker–Planck equation. Each time, it expresses the same physical fact: friction and fluctuations originate from the same bath and must be balanced so that the bath’s temperature is respected.

Example 2.4 (Two special cases)

- **Gravity**, $U = mgz$: $p^*(z) \propto e^{-mgz/T}$, the barometric profile of Perrin’s colloids.
- **Harmonic trap**, $U = \frac{1}{2}\kappa x^2$: $p^*(x) \propto e^{-\kappa x^2/2T}$, a Gaussian of variance T/κ , consistent with equipartition $\frac{1}{2}\kappa\langle x^2 \rangle = \frac{1}{2}T$. (The time-dependent problem is the Ornstein–Uhlenbeck process, whose Fokker–Planck equation $\partial_t p = k \partial_x(xp) + D \partial_x^2 p$ with $k = \kappa/\tilde{\gamma}$ has a Gaussian solution of mean $x_0 e^{-kt}$ and variance $\frac{D}{k}(1 - e^{-2kt})$.)

Remark (When can the current vanish?). The condition $\nabla \ln p^* = \mathbf{F}/(\tilde{\gamma}D)$ requires $\mathbf{F}$ to be a **gradient**: taking the curl of both sides, $\nabla \times \mathbf{F} = 0$. For a non-conservative force (a rotational field in $d \geq 2$, or a constant force on a ring) no zero-current solution exists, and any stationary state is necessarily a NESS. Nonequilibrium is therefore not only a matter of transients: some systems are driven *permanently* out of equilibrium by forces that cannot be derived from a potential, or by boundaries that impose currents.

2.7. The Chapman–Kolmogorov equation

We now derive the Fokker–Planck equation a second time, starting from the **transition probability** of a Markov process rather than from rates. This route is more general (it does not assume any specific jump mechanism) and produces naturally *two* equations, one for each end of the time interval.

Definition 2.4 (Transition probability). $W(x_0, t_0 | x, t)$ is the probability density to find the process at x at time t , given that it was at x_0 at time $t_0 \leq t$. (Time flows from left to right.) It satisfies

$$W(x_0, t_0 | x, t) \geq 0, \quad \int dx W(x_0, t_0 | x, t) = 1, \quad W(x_0, t_0 | x, t_0) = \delta(x - x_0).$$

The process is **Markovian** if, knowing the state at time s , the future after s is independent of the past before s .

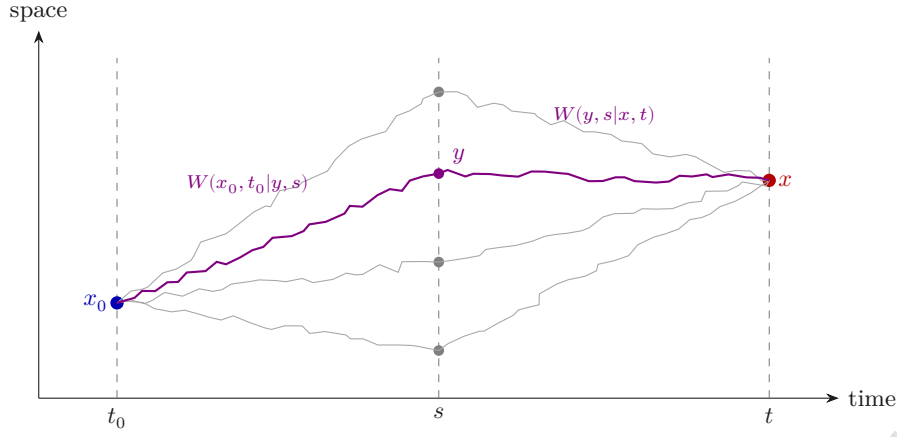
For a Markov process the following **composition law** holds for every intermediate time $s \in [t_0, t]$:

Theorem 2.6 (Chapman–Kolmogorov equation).

$$W(x_0, t_0 | x, t) = \int dy W(x_0, t_0 | y, s) W(y, s | x, t), \quad t_0 \leq s \leq t.$$

To go from (x_0, t_0) to (x, t) the process must be *somewhere*, say at y , at the intermediate time s . By the Markov property, the probability of the second leg depends on y only, not on how y was reached, so the two legs multiply; summing over all intermediate positions gives the total. This is the continuous analogue of the matrix-product rule $W^{t+t'} = W^{t'} W^t$ for Markov chains.

2. The Fokker–Planck equation



$$W(x_0, t_0 | x, t) = \int dy W(x_0, t_0 | y, s) W(y, s | x, t)$$

Figure: the Chapman–Kolmogorov equation. Every path from (x_0, t_0) to (x, t) crosses the time slice s at some position y ; for a Markov process the two legs are independent given y , and the total transition probability is the sum over all crossing points.

From the transition probability we get the evolution of any initial distribution: if $p(y, t')$ is the density at time t' ,

$$p(x, t) = \int dy p(y, t') W(y, t' | x, t), \quad t \geq t'.$$

2.8. Forward and backward Kolmogorov equations

The Chapman–Kolmogorov equation is nonlinear in W and does not, by itself, tell us how to compute W . We now exploit it for an **infinitesimal** time step, assuming (as for the master equation) that the process moves only by small amounts in short times. We define the **local drift and diffusion** of the process through the conditional moments of a short-time displacement:

$$a(y, t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int dx (x - y) W(y, t | x, t + \Delta t), \quad b^2(y, t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int dx (x - y)^2 W(y, t | x, t + \Delta t),$$

and assume that all higher conditional moments are $o(\Delta t)$ (the paths are continuous).

2.8.1. Forward equation: vary the final point

Put the short step at the **end** of the interval: $s = t$, final time $t + \Delta t$. To handle the distributional character of W , multiply by a smooth test function $f(x)$ that decays fast at infinity, and integrate:

$$\int dx f(x) \frac{\partial W(x_0, t_0 | x, t)}{\partial t} = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int dx f(x) [W(x_0, t_0 | x, t + \Delta t) - W(x_0, t_0 | x, t)].$$

By Chapman–Kolmogorov, $W(x_0, t_0 | x, t + \Delta t) = \int dy W(x_0, t_0 | y, t) W(y, t | x, t + \Delta t)$. For $\Delta t \rightarrow 0$ the short-time factor vanishes unless x is close to y , so expand $f(x) = f(y) + f'(y)(x - y) + \frac{1}{2}f''(y)(x - y)^2 + \dots$ and integrate over x . Thanks to the normalisation $\int dx W(y, t | x, t + \Delta t) = 1$, the term $f(y)$ reproduces exactly the subtracted term $\int dy f(y) W(x_0, t_0 | y, t)$ and cancels. What remains, by definition of a and b^2 , is

$$\int dx f(x) \frac{\partial W(x_0, t_0 | x, t)}{\partial t} = \int dy \left[a(y, t) f'(y) + \frac{1}{2} b^2(y, t) f''(y) \right] W(x_0, t_0 | y, t).$$

Integrating by parts (once for f' , twice for f'') moves the derivatives onto aW and b^2W ; since f is arbitrary, the integrands must coincide:

Theorem 2.7 (Forward Kolmogorov (Fokker–Planck) equation).

$$\frac{\partial W(x_0, t_0 | x, t)}{\partial t} = -\frac{\partial}{\partial x} [a(x, t) W(x_0, t_0 | x, t)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [b^2(x, t) W(x_0, t_0 | x, t)].$$

The derivatives act on the **final** variables (x, t) ; the initial point (x_0, t_0) is a fixed parameter.

This is the Fokker–Planck equation again, now for the transition probability. Comparing with the derivation from the master equation: in a short time Δt , $W(y, t | x, t + \Delta t) = [1 - \Delta t \int dz R(z, y)] \delta(x - y) + \Delta t R(x, y)$, and therefore $a = \alpha$, $b^2 = \beta$: the two definitions of drift and diffusion agree. Since $p(x, t) = \int dy p(y, t') W(y, t' | x, t)$, differentiating with respect to t shows that $p(x, t)$ obeys the same equation.

2.8.2. Backward equation: vary the initial point

Now put the short step at the **beginning**: $s = t_0 + \Delta t$,

$$W(x_0, t_0 | x, t) = \int dy W(x_0, t_0 | y, t_0 + \Delta t) W(y, t_0 + \Delta t | x, t).$$

The first factor localises y near x_0 . Expand the second factor around $y = x_0$:

$$W(y, t_0 + \Delta t | x, t) = W(x_0, t_0 + \Delta t | x, t) + (y - x_0) \partial_{x_0} W + \frac{1}{2} (y - x_0)^2 \partial_{x_0}^2 W + \dots$$

and integrate over y , using normalisation and the definitions of a , b^2 (evaluated at the starting point):

$$W(x_0, t_0 | x, t) = W(x_0, t_0 + \Delta t | x, t) + \Delta t [a(x_0, t_0) \partial_{x_0} W + \frac{1}{2} b^2(x_0, t_0) \partial_{x_0}^2 W] + o(\Delta t).$$

Dividing by Δt and letting $\Delta t \rightarrow 0$:

Theorem 2.8 (Backward Kolmogorov equation).

$$-\frac{\partial W(x_0, t_0 | x, t)}{\partial t_0} = a(x_0, t_0) \frac{\partial W(x_0, t_0 | x, t)}{\partial x_0} + \frac{1}{2} b^2(x_0, t_0) \frac{\partial^2 W(x_0, t_0 | x, t)}{\partial x_0^2}.$$

The derivatives act on the **initial** variables (x_0, t_0) ; the final point (x, t) is a fixed parameter.

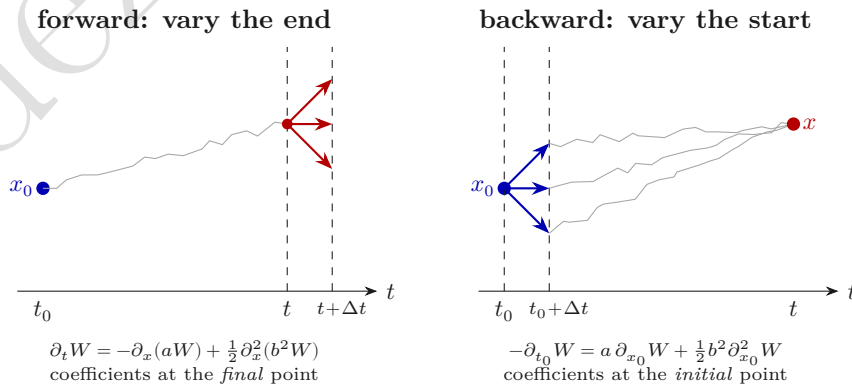


Figure: the two ways of splitting a path with Chapman–Kolmogorov. Forward: the last infinitesimal step is expanded, and the equation describes how the distribution of the endpoint evolves. Backward: the first infinitesimal step is expanded, and the equation describes how the transition probability depends on where the process starts.

2. The Fokker–Planck equation

Remark (Forward and backward are adjoint). The two equations are **not** symmetric: in the forward one the coefficients sit *inside* the derivatives, in the backward one *outside*. Defining

$$\mathcal{L}g = -\partial_x(ag) + \frac{1}{2}\partial_x^2(b^2g), \quad \mathcal{L}^\dagger g = a\partial_x g + \frac{1}{2}b^2\partial_x^2g,$$

one has $\int dx h \mathcal{L}g = \int dx g \mathcal{L}^\dagger h$ (integration by parts): $\mathcal{L}^\dagger$ is the adjoint of the Fokker–Planck operator $\mathcal{L}$. The forward equation is $\partial_t W = \mathcal{L}_x W$, the backward one is $-\partial_{t_0} W = \mathcal{L}_{x_0}^\dagger W$. The adjoint operator is the one that appeared in the test-function step: $\frac{d}{dt}\langle f(x_t) \rangle = \langle \mathcal{L}^\dagger f(x_t) \rangle$.

For a **time-homogeneous** process (a, b independent of t), W depends on $t - t_0$ only, hence $\partial_{t_0} W = -\partial_t W$ and the backward equation reads $\partial_t W = a(x_0)\partial_{x_0} W + \frac{1}{2}b^2(x_0)\partial_{x_0}^2 W$. As a check, for drift-diffusion with constant $a = v_0$, $b^2 = 2D$, the Gaussian propagator depends on $x - x_0 - v_0 t$, so $\partial_{x_0} = -\partial_x$ on it: forward and backward equations are then the same equation.

Remark (What the backward equation is for). The backward equation is the natural tool whenever the question is “starting from x_0 , what happens?”: the probability to reach one boundary before another, the distribution and the mean of the **first exit time** from a domain, escape over a barrier (Kramers’ and Arrhenius’ formulas). For example, the probability $\pi(x_0)$ to be absorbed at x_2 before x_1 obeys the stationary backward equation $a(x_0)\pi' + \frac{1}{2}b^2(x_0)\pi'' = 0$ with $\pi(x_1) = 0$, $\pi(x_2) = 1$; for pure diffusion $\pi(x_0) = (x_0 - x_1)/(x_2 - x_1)$, in agreement with the source-and-sinks example above.

2.9. Summary: one process, several descriptions

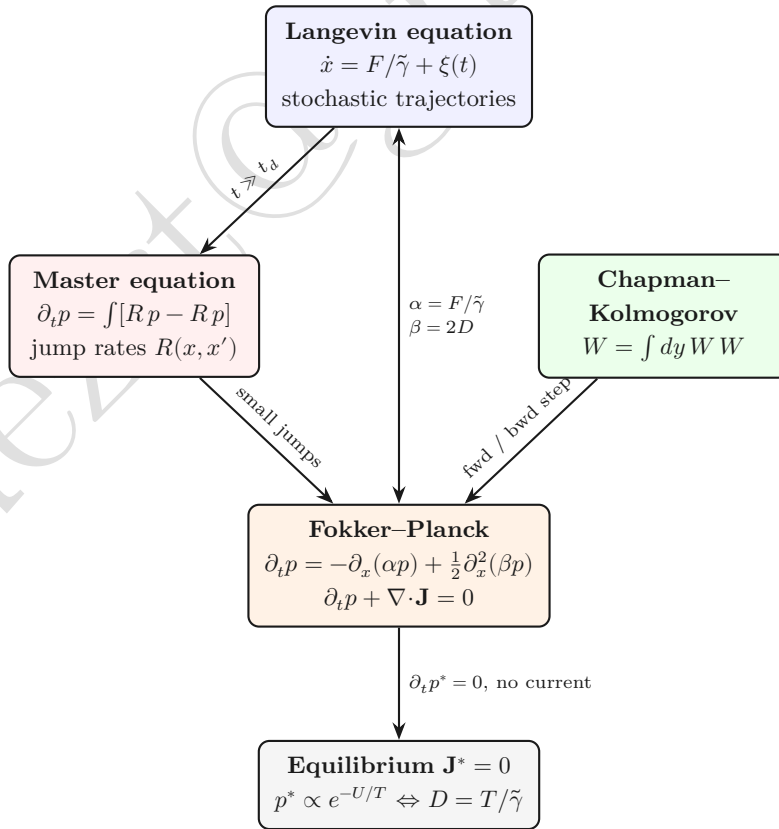


Figure: the descriptions of a Markovian diffusion process met in Chapters 1–2 and how they are connected.

- The master equation reduces to the Fokker–Planck equation when jumps are small compared with

the scale of variation of p ; only the first two jump moments survive, the **drift velocity** α and the **diffusion** β ($= 2D$ for an isotropic medium).

- Without drift the propagator is a Gaussian of variance $2Dt$ (per dimension), which spreads as $\sqrt{t}$; a drift translates it at constant velocity.
- The Fokker–Planck equation is a continuity equation, $\partial_t p + \nabla \cdot \mathbf{J} = 0$, with $\mathbf{J} = \alpha p - \frac{1}{2} \nabla \cdot (\beta p)$ (drift current plus Fick current).
- **Stationary** means $\nabla \cdot \mathbf{J}^* = 0$; **equilibrium** means $\mathbf{J}^* = 0$ (the continuum version of detailed balance). Stationary states with $\mathbf{J}^* \neq 0$ are NESS and require non-conservative forces, periodic geometries or driving boundaries.
- For a conservative force the zero-current state is $p^* \propto e^{-U/(\tilde{\gamma}D)}$; identifying it with Boltzmann gives the Einstein relation $D = T/\tilde{\gamma}$.
- For a Markov process the transition probability obeys the Chapman–Kolmogorov equation; expanding the last (first) short step gives the forward (backward) Kolmogorov equation, two mutually adjoint forms of the same dynamics.

2.10. Exercises

Exercise 2.1

(Checking the propagator) Verify by direct differentiation that

$$p(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp \left[-\frac{(x - x_0 - v_0 t)^2}{4Dt} \right]$$

solves $\partial_t p = -v_0 \partial_x p + D \partial_x^2 p$, and that $p(x, t) \rightarrow \delta(x - x_0)$ as $t \rightarrow 0^+$.

Solution

With $u = x - x_0 - v_0 t$: $\partial_x p = -\frac{u}{2Dt} p$, $\partial_x^2 p = (\frac{u^2}{4D^2 t^2} - \frac{1}{2Dt}) p$, $\partial_t p = (-\frac{1}{2t} + \frac{uv_0}{2Dt} + \frac{u^2}{4Dt^2}) p$. Then $-v_0 \partial_x p + D \partial_x^2 p = (\frac{v_0 u}{2Dt} + \frac{u^2}{4Dt^2} - \frac{1}{2t}) p = \partial_t p$. As $t \rightarrow 0^+$ the Gaussian has unit area and variance $2Dt \rightarrow 0$, so it converges to $\delta(x - x_0)$.

Exercise 2.2

(Kramers–Moyal for a biased walk) For the lattice walk with $R(x; \chi) = r_+ \delta(\chi - a) + r_- \delta(\chi + a)$, compute all the jump moments M_n . Show that in the scaling limit $a \rightarrow 0$ with $(r_+ - r_-)a = v$ and $(r_+ + r_-)a^2 = 2D$ fixed, all $M_{n \geq 3}$ vanish. What happens if instead $r_+ - r_-$ stays finite as $a \rightarrow 0$?

Solution

$M_n = a^n [r_+ + (-1)^n r_-]$: $M_{2m} = a^{2m} (r_+ + r_-) = 2D a^{2m-2} \rightarrow 0$ for $m \geq 2$; $M_{2m+1} = a^{2m+1} (r_+ - r_-) = v a^{2m} \rightarrow 0$ for $m \geq 1$. If $r_+ - r_-$ stays finite while $(r_+ + r_-)a^2$ is fixed, then $v = (r_+ - r_-)a \rightarrow 0$: the drift disappears in the diffusive limit. A drift survives only if the bias scales as $r_+ - r_- \sim 1/a$, i.e. it is weak compared with the total rate $\sim 1/a^2$.

Exercise 2.3

(Ornstein–Uhlenbeck process) For $\partial_t p = k \partial_x (xp) + D \partial_x^2 p$, derive the equations for $\langle x \rangle(t)$ and $\langle x^2 \rangle(t)$, solve them with $p(x, 0) = \delta(x - x_0)$, and show that the variance is $\frac{D}{k}(1 - e^{-2kt})$. Find the stationary distribution by imposing $J^* = 0$, and show that it is consistent with Boltzmann for $U = \frac{1}{2} \kappa x^2$ if $k = \kappa/\tilde{\gamma}$ and $D = T/\tilde{\gamma}$.

Exercise 2.4

(Equilibrium is impossible on a tilted ring) On a ring of length L consider the force $F(x) = F_0 - U'(x)$ with U periodic. Show that the stationary current is

$$J^* = \frac{D(1 - e^{-F_0 L/T})}{\int_0^L dx \int_0^L dy e^{[V(x) - V(x-y)]/T}}, \quad V(x) = U(x) - F_0 x,$$

(with V extended by $V(x - L) = V(x) + F_0 L$), and in particular that $J^* = 0$ if and only if $F_0 = 0$. Check that for $U = 0$ you recover $J^* = F_0/(\zeta L)$.

Exercise 2.5

(Splitting probability from the backward equation) A particle diffuses with constant drift $v_0 > 0$ and diffusion coefficient D in $[0, L]$, with absorbing ends. Using the stationary backward equation, find the probability $\pi(x_0)$ that a particle starting at x_0 exits through L . Discuss the limits $v_0 L/D \rightarrow 0$ and $v_0 L/D \rightarrow \infty$.

Solution

$v_0 \pi' + D \pi'' = 0$ with $\pi(0) = 0$, $\pi(L) = 1$ gives $\pi(x_0) = \frac{1 - e^{-v_0 x_0/D}}{1 - e^{-v_0 L/D}}$. For $v_0 L/D \rightarrow 0$ (diffusion dominates) $\pi \rightarrow x_0/L$, the result for pure diffusion. For $v_0 L/D \rightarrow \infty$ (drift dominates) $\pi \rightarrow 1$ for any $x_0 \gg D/v_0$: only particles starting within a boundary layer of thickness D/v_0 near 0 have an appreciable chance to exit on the left. The dimensionless number $v_0 L/D$ is a Péclet number.

Exercise 2.6

(Mean exit time) Show that the mean exit time $\mathcal{T}(x_0)$ from an interval obeys $a(x_0)\mathcal{T}' + \frac{1}{2}b^2(x_0)\mathcal{T}'' = -1$, by integrating the backward equation for the survival probability $\mathcal{P}_{x_0}(t) = \int_I dx W(x_0, 0|x, t)$ over time. Solve it for pure diffusion with absorbing ends and compare with the source-and-sinks result $\frac{(x_0 - x_1)(x_2 - x_0)}{2D}$.

2.11. Sources

Lecture 2 of the course: derivation of the Fokker–Planck equation from the master equation in the general case, with drift velocity and diffusion coefficient; Gaussian solution of the diffusion equation; the Fokker–Planck equation as a continuity equation with the probability current in d dimensions; stationarity versus macroscopic equilibrium (no current); the equilibrium distribution for a conservative force and the Einstein relation from the Boltzmann distribution; the continuous Chapman–Kolmogorov equation and the forward and backward Kolmogorov equations. Book: R. Livi and P. Politi, *Nonequilibrium Statistical Physics: A Modern Perspective*, Cambridge (2017), §1.4.2 (with $W \rightarrow R$), beginning and last subsection of §1.6.3, first part of §1.6.4; the discrete global and detailed balance conditions anticipate §1.5.4, the Ornstein–Uhlenbeck example and the source-and-sinks example are from §1.6.3, and the backward derivation by first-step splitting complements Appendix E (Kramers–Moyal) of the book.